307.504 Aircraft Design I (WS2025)

Preliminary Design Specification of a commercial aircraft with electric propulsion system, LH2 and fuel cell

Lena Aveline, Benoit Pereira, Michael Riedl, Matthias Frenzl

Final Report

January 19, 2026

Contents

I Design Basis **5**

1 Introduction **5**

2 Requirements (High Level) **6**
2.1 Regulatory Requirements . . . 6
2.2 Baseline Requirements . . . 7
2.3 Stretched Variant Requirements . . . 10
2.4 Assumptions . . . 13

II Initial Mass Estimation **14**

3 Base Variant **14**
3.1 Objectives & Requirements . . . 14
3.1.1 Operating empty mass payload, fuel mass, take-off mass . . . 14
3.1.2 Payload factors . . . 14
3.1.3 Fuel . . . 14
3.1.4 Specifications . . . 15
3.2 Methods . . . 15
3.2.1 Process . . . 15
3.2.2 Physical range . . . 15
3.2.3 Database charts . . . 16
3.2.4 First mass calculations . . . 16
3.2.5 Correction for technological advancements . . . 17
3.2.6 Correction LH2 Propulsion . . . 17
3.3 Results . . . 18
3.3.1 Baseline Masses . . . 18

4 Stretched Variant **19**
4.1 Objectives & Requirements . . . 19
4.1.1 New Specifications . . . 19
4.1.2 Commonalities . . . 19
4.2 Methods . . . 19
4.2.1 Differences . . . 19
4.2.2 Process . . . 19
4.3 Results . . . 20
4.3.1 Variant Masses . . . 20

5 Discussion about commonalities **21**

III Initial Propulsive Performance Assessment **22**

6 Methods **22**
6.1 Estimate the power required for the aircraft . . . 22
6.2 Approach speed and landing Requirements . . . 22
6.3 Shaft Power . . . 22

7 Assumptions **22**

8 Results **24**
8.1 Chart P/W vs W/S . . . 24
8.2 Approach speed and landing requirement . . . 26
8.3 Shaft Power . . . 27
8.4 Propeller Diameter . . . 27

IV Hydrogen systems **29**

9 HEF System design choices **29**
9.1 Architecture layout . . . 29
9.2 Fuel cells load optimization . . . 30

10 HEF System Volume & Mass Estimation **31**
10.1 Methods . . . 31
10.1.1 Hydrogen Tanks . . . 31
10.1.2 Fuel Cells . . . 31
10.1.3 Batteries . . . 32
10.1.4 Inverters . . . 32
10.1.5 Electric Motors . . . 32
10.2 Results . . . 32

V Configuration **33**

11 Fuselage **33**
11.1 Methods & Assumptions . . . 33
11.2 Results . . . 34

12 Wing **37**
12.1 Methods . . . 37
12.1.1 Lift Coefficient . . . 37
12.1.2 Sweep . . . 38
12.1.3 Geometry . . . 39
12.2 Results . . . 40
12.2.1 Lift coefficients . . . 40
12.2.2 Geometry . . . 40

13 Empennage **41**
13.1 Methods 41
13.2 Results 41

14 HEF System integration **42**

VI Conclusion **44**

15 Design Summary **44**

16 Requirement Compliance **45**

17 Family Concept Assessment **45**

VII Appendix A **47**

18 Used Python Code **47**
18.1 First mass estimation - Values at point B 47
18.2 First mass estimation - Tanks commonality 51
18.3 Initial propulsion system sizing - Wing Loading - Baseline 53
18.4 Initial propulsion system sizing - Wing Loading - Stretch Variant 56
18.5 Landing Requirements and Propeller Dimensioning 59
18.6 Commonalities impacts 61
18.7 Hydrogen Systems - Fuel cells load optimizer 62
18.8 Hydrogen Systems - Mass and volume estimation 66
18.9 Hydrogen Systems - CAD help 70
18.10 Configuration - Wing Dimensioning 73
18.11 Configuration - Empennage Dimensioning 78
18.12 Configuration - Fuselage Dimensioning 79

Part I
Design Basis

1 Introduction

To reduce aviation's reliance on fossil fuels and to decrease its impacts on the environment, airplanes manufacturers experiment with new kinds of propulsion systems. Among them, electric propulsion powered by hydrogen through fuel cells seems to have a huge potential. Our goal in this course is to learn the basics of aircraft design and apply them in a 4-person project related to the actuality of hydrogen powered aircrafts. We will conduct the conceptual design phase for a commercial aircraft with liquid hydrogen power, fuel cells and electric propulsion.

Our aircraft will be able to carry 48 passengers [RQ-01] and 500kg [RQ-04] of freight over 700NM (1300km) [RQ-07] at an altitude of at least 25000ft (7600m) [RQ-14] and above Mach 0.5 [RQ-12] We will also develop the family concept by including in our design a stretch variant that should be able to carry 70 passengers [RQ-26]. Both variants are compliant to the according regulations [RQ-17]ff, like EASA Air-Operations ANNEX IV (PART-CAT, SUBPART B: OPERATING PROCEDURE)[1] and CS36[2] among others.

To guide us we will follow the lecture methods and guidelines, such as the EASA certification guidelines. We will also use the database [3] at our disposition and compare our results against planes with similar specifications.
This report aims to present and detail the work done during the semester, from the initial mass estimation to the final drag estimation results.

2 Requirements (High Level)

2.1 Regulatory Requirements

#	Type	Title	Author	Date
RegRQ-1	Certification Specifications	Certification Specifications and Acceptable Means of Compliance for Large Aeroplanes (CS-25) Amendment 28	EASA	2023
RegRQ-2	Regulation	Air-Operations (IR + AMC)	EASA	February 2025
RegRQ-3	Regulation	CS-36 Amendment 6	EASA	2021
RegRQ-4	Regulation	Annex 16 to the Convention on International Civil Aviation, Environmental Protection. Volume 1 Aircraft Noise, 8th Edition	ICAO	2017
RegRQ-5	Regulation	UK Regulation (EU) No. 965/2012. Air Operations - Annex III (Part-ORO), ORO.CC.100 Number and composition of cabin crew	Civil Aviation Authority	2025
RegRQ-6	Regulation	CS 25.803 Emergency evacuation	EASA	2023
RegRQ-7	Regulation	CS 25.807 Emergency exits	EASA	2023

Table 1: Regulatory Requirements

2.2 Baseline Requirements

Table 2: Baseline Requirements

Requ. ID	Layer	Variant	Mandatory?	Category	Short	Long	Value	Unit	Ref.	Notes
RQ-01	1	Baseline	Shall	Passenger	Passenger capacity	Absolute Passenger Number	48	Pax	[4]	
RQ-02	1	Baseline	Shall	Passenger	Mass per passenger	Avg Mass per passenger	83	kg	[4]	
RQ-03	1	Baseline	Shall	Passenger	Luggage per passenger	Avg Luggage per passenger	12	kg	[4]	
RQ-04	1	Baseline	Shall	Payload	Additional payload capacity		500	kg	[4]	
RQ-05	1	Baseline	Shall	Distance	Take-off distance	Max take-off distance at max take-off mass	1300	m	[4]	SL, ISA conditions
RQ-06	1	Baseline	Shall	Distance	Landing distance	Max landing distance	1250	m	[4]	SL, ISA conditions
RQ-07	1	Baseline	Shall	Range	Operational Range	Operational Range with maximum payload and standard fuel reserves	700	NM	[4]	
RQ-08	1	Baseline	Shall	Fuel reserves	Standard fuel reserves	Operational range shall be achieved with standard fuel reserves			[4]	5% of cruise level
RQ-09	1	Baseline	Shall	Fuel reserves	Standard fuel reserves	Operational range shall be achieved with standard fuel reserves			[4]	100 NM alternate flight at FL 200
RQ-10	1	Baseline	Shall	Fuel reserves	Standard fuel reserves	Operational range shall be achieved with standard fuel reserves			[4]	0.5h holding
RQ-11	1	Baseline	Shall	Velocity	Approach speed	Max approach speed at max landing mass	120	kts	[4]	
RQ-12	1	Baseline	Shall	Velocity	Mach number	Min mach number	0.5		[4]	
RQ-13	1	Baseline	Shall	Altitude	One-engine inoperative	Minimum Altitude after one-engine failure	10,000	ft	[4]	
RQ-14	1	Baseline	Shall	Altitude	Service ceiling	Minimum service ceiling with residual climb > 300ft/min	25,000	ft	[4]	
RQ-15	2	Baseline	Should	Structure	Materials	Different structure Materials	Composite / metal			
RQ-16	1	Baseline	Shall	propulsion	Power Source	Propulsion based on hydrogen fuel cell and electric motor			[4]	

Requ. ID	Layer	Variant	Mandatory?	Category	Short	Long	Value	Unit	Ref.	Notes
RQ-17	1	Baseline	Shall	Weight	Weight limits	Maximum and minimum aircraft weights must ensure compliance with all structural, performance and noise requirements			[5] CS25.25 p.98-99	
RQ-18	1	Baseline	Shall	Take-off	Take-off speeds	V1, V2, VR must be established following CS25.107 definitions			[5] CS25.107 p.110	
RQ-19	1	Baseline	Shall	Take-off	Take-off path	Aircraft must comply with all-engine and OEI climb gradients and obstacle clearance			[5] CS25.111 p.128	
RQ-20	1	Baseline	Shall	Take-off	Take-off distance	Take-off run and distance in wet and dry runway must be computed according to CS25.113			[5] CS25.113 p.130	
RQ-21	1	Baseline	Shall	Take-off	Take-off flight path	Aircraft must satisfy CS25.115 take-off flight path requirements			[5] CS25.115 p.131	
RQ-22	1	Baseline	Shall	Climb	One-engine inoperative climb	One-engine inoperative climb gradients must satisfy CS25.121			[5] CS25.121 p.133	
RQ-23	1	Baseline	Shall	Landing	Landing Distance	Landing distance must comply with CS25.125			[5] CS25.125 p.136	
RQ-24	1	Baseline	Shall	Safety	Emergency exits	Full aircraft evacuation within a certain time. Regulation CS 25.803	90	s	[5] CS25.803 (c)	
RQ-25	1	Baseline	Shall	Safety	Emergency exits	Limitation on emergency exits used during evacuation demonstration Regulation CS 25.803			[5] CS25.803 Appendix J (p)	
RQ-26	1	Baseline	Shall	Safety	Emergency exits capacity	Passenger seating limits associated with Type III emergency exits. Regulation CS 25.807 (g)			[5] CS25.807 (g)-(7)	
RQ-27	1	Baseline	Shall	Safety	Emergency exits configuration	Minimum number of large emergency exits per side of the fuselage Regulation CS 25.807 (g)	2		[5] CS25.807 (g)-(8)	
RQ-28	1	Baseline	Shall	Safety	Alternative Emergency exits	Egress performance of ventral or tail cone exits. Regulation CS 25.807 (g)			[5] CS25.807 (g)-(9)	

Requ. ID	Layer	Variant	Mandatory?	Category	Short	Long	Value	Unit	Ref.	Notes
RQ-29	1	Baseline	Shall	Safety	Cabin surveillance	Minimum passenger visibility per cabin compartment	50	%	[6]	
RQ-30	1	Baseline	Shall	Safety	Cabin surveillance	Overall passenger visibility by flight attendants	80	%	[6]	
RQ-31	1	Baseline	Shall	Operation &Safety	Cabin crew complement	Minimum number of flight attendants	1 for 50 passengers		[6]	
RQ-32	1	Baseline	Shall	Cabin layout	Seating arrangement	Maximum number of seats abreast in single-aisle aeroplanes. Regulation CS-25.817			[5] CS25.817	
RQ-33	1	Baseline	Shall	Safety	Cabin Interior	Lavatory doors must prevent passenger entrapment. Regulation CS-25.820			[5]CS-25.820	
RQ-34	1	Baseline	Shall	Passenger	Seat Class	Number of comfort and price classes	1		[4]	

2.3 Stretched Variant Requirements

Table 3: Stretch Variant Requirements

Requ. ID	Layer	Variant	Mandatory?	Category	Short	Long	Value	Unit	Ref.	Notes
RQ-35	1	Stretch	Shall	Passenger	Passenger capacity	Absolute Passenger Number	70	Pax	[4]	
RQ-36	1	Stretch	Shall	Passenger	Mass per passenger	Avg Mass per passenger	83	kg	[4]	
RQ-37	1	Stretch	Shall	Passenger	Luggage per passenger	Avg Luggage per passenger	12	kg	[4]	
RQ-38	1	Stretch	Shall	Payload	Additional payload capacity		500	kg	[4]	
RQ-39	1	Stretch	Shall	Distance	Take-off distance	Max take-off distance at max take-off mass	1300	m	[4]	SL, ISA conditions
RQ-40	1	Stretch	Shall	Distance	Landing distance	Max landing distance	1250	m	[4]	SL, ISA conditions
RQ-41	1	Stretch	Shall	Range	Operational Range	Operational Range with maximum payload and standard fuel reserves	700	NM	[4]	
RQ-42	1	Stretch	Shall	Fuel reserves	Standard fuel reserves	Operational range shall be achieved with standard fuel reserves			[4]	5% of cruise level
RQ-43	1	Stretch	Shall	Fuel reserves	Standard fuel reserves	Operational range shall be achieved with standard fuel reserves			[4]	100 NM alternate flight at FL 200
RQ-44	1	Stretch	Shall	Fuel reserves	Standard fuel reserves	Operational range shall be achieved with standard fuel reserves			[4]	0.5h holding
RQ-45	1	Stretch	Shall	Velocity	Approach speed	Max approach speed at max landing mass	120	kts	[4]	
RQ-46	1	Stretch	Shall	Velocity	Mach number	Min mach number	0.5		[4]	
RQ-47	1	Stretch	Shall	Altitude	One-engine inoperative	Minimum Altitude after one-engine failure	10,000	ft	[4]	
RQ-48	1	Stretch	Shall	Altitude	Service ceiling	Minimum service ceiling with residual climb > 300ft/min	25,000	ft	[4]	
RQ-49	2	Stretch	Should	Structure	Materials	Different structure Materials	Composite/ metal			
RQ-50	1	Stretch	Shall	Propulsion	Power Source	Propulsion based on hydrogen fuel cell and electric motor			[4]	

Requ. ID	Layer	Variant	Mandatory?	Category	Short	Long	Value	Unit	Ref.	Notes
RQ-51	1	Stretch	Shall	Weight	Weight limits	Maximum and minimum aircraft weights must ensure compliance with all structural, performance and noise requirements			[5] CS25.25 p.98-99	
RQ-52	1	Stretch	Shall	Take-off	Take-off speeds	V1, V2, VR must be established following CS25.107 definitions			[5] CS25.107 p.110	
RQ-53	1	Stretch	Shall	Take-off	Take-off path	Aircraft must comply with all-engine and OEI climb gradients and obstacle clearance			[5] CS25.111 p.128	
RQ-54	1	Stretch	Shall	Take-off	Take-off distance	Take-off run and distance in wet and dry runway must be computed according to CS25.113			[5] CS25.113 p.130	
RQ-55	1	Stretch	Shall	Take-off	Take-off flight path	Aircraft must satisfy CS25.115 take-off flight path requirements			[5] CS25.115 p.131	
RQ-56	1	Stretch	Shall	Climb	One-engine inoperative climb	One-engine inoperative climb gradients must satisfy CS25.121			[5] CS25.121 p.133	
RQ-57	1	Stretch	Shall	Landing	Landing Distance	Landing distance must comply with CS25.125			[5] CS25.125 p.136	
RQ-58	1	Stretch	Shall	Safety	Emergency exits	Full aircraft evacuation within a certain time. Regulation CS 25.803	90	s	[5] CS25.803 (c)	
RQ-59	1	Stretch	Shall	Safety	Emergency exits	Limitation on emergency exits used during evacuation demonstration Regulation CS 25.803			[5] CS25.803 Appendix J (p)	
RQ-60	1	Stretch	Shall	Safety	Emergency exits capacity	Passenger seating limits associated with Type III emergency exits. Regulation CS 25.807 (g)			[5] CS25.807 (g)-(7)	
RQ-61	1	Stretch	Shall	Safety	Emergency exits configuration	Minimum number of large emergency exits per side of the fuselage Regulation CS 25.807 (g)	2		[5] CS25.807 (g)-(8)	
RQ-62	1	Stretch	Shall	Safety	Alternative Emergency exits	Egress performance of ventral or tail cone exits. Regulation CS 25.807 (g)			[5] CS25.807 (g)-(9)	

Requ. ID	Layer	Variant	Mandatory?	Category	Short	Long	Value	Unit	Ref.	Notes
RQ-63	1	Stretch	Shall	Safety	Cabin surveillance	Minimum passenger visibility per cabin compartment	50	%	[6]	
RQ-64	1	Stretch	Shall	Safety	Cabin surveillance	Overall passenger visibility by flight attendants	80	%	[6]	
RQ-65	1	Stretch	Shall	Operation &Safety	Cabin crew complement	Minimum number of flight attendants	1 for 50 passengers		[6]	
RQ-66	1	Stretch	Shall	Cabin layout	Seating arrangement	Maximum number of seats abreast in single-aisle aeroplanes. Regulation CS-25.817			[5] CS25.817	
RQ-67	1	Stretch	Shall	Safety	Cabin Interior	Lavatory doors must prevent passenger entrapment. Regulation CS-25.820			[5]CS-25.820	
RQ-68	1	Stretch	Shall	Passenger	Seat Class	Number of comfort and price classes	1		[4]	

2.4 Assumptions

Table 4: Assumptions overview

Assump. ID	Type	Description
ASS-01	Weather condition	We assume that the aircraft is operating in ISA (International Standard Atmosphere) conditions, i.e. in a standardized atmosphere characterised by a temperature of 15°C and a pressure of 1013.25 hPa at sea level, as well as density and temperature values that theoretically vary with altitude.
ASS-02	Scaling factor	We assume a scaling factor of 0.4 to represent the increase in operating mass for a given increase in payload mass.
ASS-03	Relationship between physical range and fuel mass	We assume that the physical range of the aircraft is proportional to its fuel mass.
ASS-04	Values for propulsion	We assume a glide ratio $\varepsilon = 20$, which takes further development of the aerodynamics into account.
ASS-05	Wing - Specific Heats Ratio	The ratio of specific heats was assumed equal to $\gamma = 1.4$, corresponding to dry air under standard atmospheric conditions
ASS-06	Wing - Aspect Ratio	The wing aspect ratio was assumed equal to AR=8 during the initial propulsion system sizing.
ASS-07	Wing - Taper Ratio	The wing taper ratio was selected equal to that of the ATR-42 reference aircraft. $\lambda = 0.45$
ASS-08	Gravimetric Efficiency	We assume an average gravimetric efficiency $GE = 50\%$
ASS-09	Throttle Ratio - Cruise	We assume a throttle ratio of 0.8 during cruise
ASS-10	Throttle Ratio - Climb	We assume a throttle ratio of 0.9 during climb
ASS-11	Throttle Ratio - Take-Off	We assume a throttle ratio of 1 during take-off
ASS-12	FC ancillaries self-consumption	We assume a 7% self-consumption of the FC ancillaries, disregarding weather conditions
ASS-13	Fuel Factor	For our aircraft we assumed a 20% decrease of the fuel factor, reflecting better engine efficiency
ASS-14	Payload Factor	Taking into account the rapid development of new materials we assumed a 10% increase in payload factor, reflecting a lighter aircraft with the same capacity

Part II
Initial Mass Estimation

3 Base Variant

3.1 Objectives & Requirements

3.1.1 Operating empty mass payload, fuel mass, take-off mass

The "total" mass of an aircraft is called the take-off mass. This is the mass that has to be lifted by the wings and accelerated by the engines.

This mass can be subdivided in three:

1. the payload mass includes the passengers, their luggage and the freight
2. the fuel mass
3. the operating-empty mass which corresponds to the mass of all systems that are necessary for the flight. This is roughly the mass of the airframe plus the mass of the crew, equipments and accessories

These characteristic masses are very important parameters that we need for the rest of our project. The first step of our conceptual design will then consist of a first mass estimation of our aircraft.

3.1.2 Payload factors

To evaluate the values of the fuel mass and the payload mass to the take-off mass, we use respectively the payload factor and the fuel factor. Such factors are interesting to evaluate the design of an aircraft : if the fuel factor is high this means that the airplane tends to become a giant fuel tank which is not interesting economically. On the contrary if the payload factor is high it means that a large fraction of the mass carried by the aircraft is the payload, which is economically desirable.

3.1.3 Fuel

To fly our aircraft safely we have to bring fuel for the trip but also some reserves which will increase our fuel mass. There are different reserves for different situations. In case our fuel consumption exceeds the calculated one (for example in case of headwind), we have a contingency reserve. In case we need to land at another airport we have a diversion reserve. Finally, if our landing is delayed at the airport we need a holding reserve. In normal flight conditions such reserves should not be needed.

3.1.4 Specifications

Our mass estimation will start with the baseline. For these calculations some specifications will be very important : the range and fuel reserves of our aircraft as well as the parameters of the payload : number of passengers, mass of freight, luggage, ...

3.2 Methods

3.2.1 Process

To obtain the characteristic masses of our baseline aircraft, we will use the following method:

1. Physical and operational range calculations
2. Estimation of our payload and fuel factors, by comparison with aircrafts having a similar range to ours in the database.
3. First calculation of the characteristic masses
4. Correction for future technological advancements expected for 2040
5. Correction for electric-LH2 propulsion system

3.2.2 Physical range

The first thing we have to compute is the physical range of our aircraft. It is the sum of the operative range and the reserves ranges. We find all the necessary informations in the assignment specification [4]:

- The operative range and diversion reserve are directly available
- The contingency reserve is proportional to the operative range, in our case 5% [RQ-08]
- The holding reserve is the simple product of the holding time by the holding speed : 30minutes at 200knots [RQ-10]

Then we need to compute the operational range of the database aircrafts. For this we use the following formulas:

$$\frac{dm_F}{dr} = \frac{m_{TO} - m_{OE} - m_{Pc}}{r_{PH}} \tag{Fuel Gradient}$$

$$r_{OP} = r_{PH} + \frac{m_{OE} + m_{Pc} - m_{ZF}}{\frac{dm_F}{dr}} \tag{Op. Range}$$

3.2.3 Database charts

We can now calculate the payload and fuel factor for all aircrafts of the database and plot them on a diagram against the operational range we just computed.
We then apply a linear regression on both point clouds. With this we are able to compare our aircraft to those in the database [3] and we can estimate what our fuel and payload factors should be for the baseline.

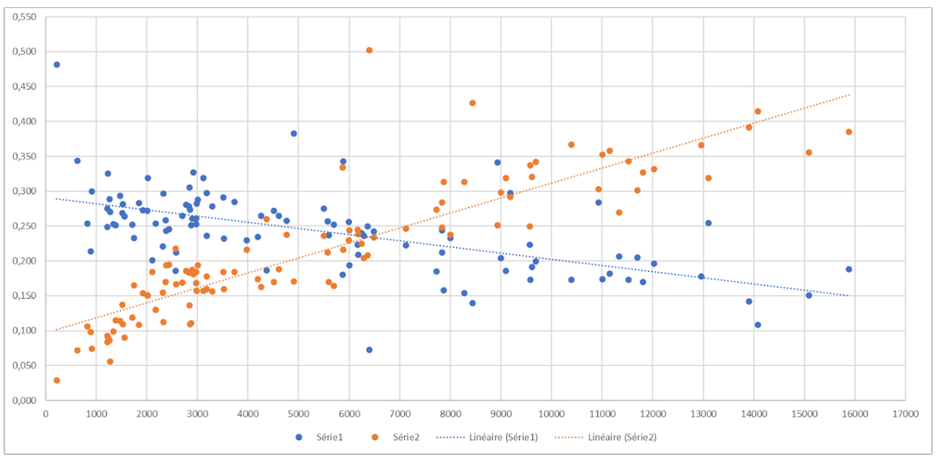

Figure 1: Baseline vs Database [3]

3.2.4 First mass calculations

With the payload and fuel factors we can easily compute the characteristic masses of our baseline:

- The payload mass can easily be calculated from the specifications : we multiply the assumed mass of passengers (with their luggage) by the number of passengers and add the 500kg of freight:

$$m_{P,bas} = N_{bPass1} \cdot (m_{Pass} + m_{Lugg}) + m_{Frght} \qquad \text{(Payload Mass)}$$

- The maximum take-off mass is obtained by dividing the payload mass by the payload factor.

$$m_{TO,bas} = \frac{m_{P,bas}}{y} \qquad \text{(MTOM)}$$

- The fuel mass is obtained by multiplying the fuel factor and the maximum take-off mass.

$$m_{F,bas} = k \cdot m_{TO,bas} \qquad \text{(Fuel Mass)}$$

- The operating-empty mass is obtained by simple subtraction.

$$m_{OE,bas} = m_{TO,bas} - m_{P,bas} - m_{F,bas} \qquad \text{(OEM)}$$

3.2.5 Correction for technological advancements

The certification and production of our aircraft is not expected to be finished before 2040, assuming a design freeze in 2030. By this time we can expect some improvements in engine efficiency and in the airframe design. To take into account those improvements, we chose to correct our payload and fuel factors.

For our aircraft we assumed a 20% decrease of the fuel factor[ASS-13], reflecting better engine efficiency.

Taking into account the rapid development of new materials we assumed a 10% [ASS-14] increase in payload factor, reflecting a lighter aircraft with the same capacity. With these new factors we can compute the characteristic masses again.

3.2.6 Correction LH2 Propulsion

Until now our calculations have been based on kerosene powered aircraft. However the main characteristic of our aircraft is its electric-LH2 propulsion which creates differences at multiple levels:

- the propulsion system has to be replaced with some components discarded and other added
- hydrogen has higher gravimetrical energy density [7] and allows a better drive efficiency but evaporates and requires bigger tanks that increase drag/

To take these differences into account we implemented new corrective coefficients to the fuel factor and the operating empty mass, with repercussions on the fuel and maximum take-off mass.

3.3 Results

3.3.1 Baseline Masses

Estimated characteristic masses of the baseline at design point:

Parameter	**Baseline Value**
Operational Range	700 NM
Physical Range	935 NM
Operating Empty Mass (m_{OE})	12,728 kg
Payload Mass (m_{PL})	5,060 kg
Fuel Mass (m_{fuel})	572 kg
Maximum Take-off Mass (m_{MTO})	18,360 kg

Table 5: Baseline Mass

Payload-range diagram for the baseline aircraft:

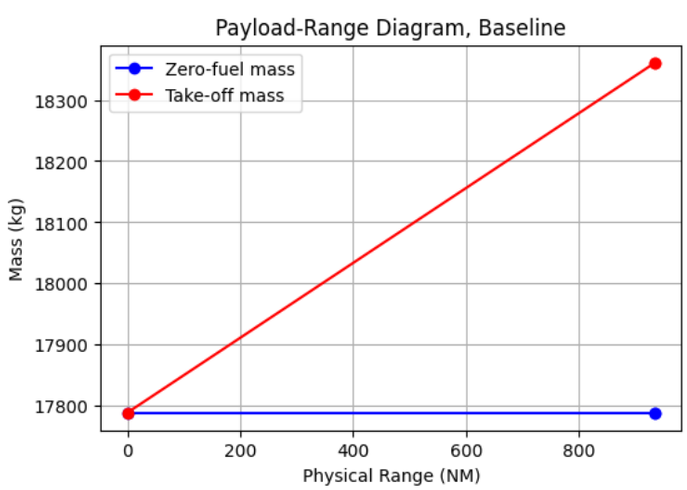

Figure 2: Baseline

4 Stretched Variant

4.1 Objectives & Requirements

4.1.1 New Specifications

To respond to the diversity of airliners demands, aircraft manufacturers often develop the family concept [8]. The idea is to modify an existing airplane in order to meet new requirements without designing a new aircraft from scratch, thus reducing costs for both the manufacturer and the customer.

For our project this "stretched" variant differs from the baseline by carrying 70 passengers [RQ-07] instead of 48 [RQ-01] remaining all other requirements unchanged.

4.1.2 Commonalities

Our goal is to reuse as much hardware as possible : wings, engines, cockpit, control surfaces and the LH2 tanks. In this way, we will only stretch the fuselage of our baseline to create this new stretched variant.

4.2 Methods

4.2.1 Differences

Our first design idea has been to iterate our previous calculations with the new number of passengers.
However this would have resulted in an overall bigger aircraft that doesn't fit the concept of a stretched variant and doesn't allow us to reuse hardware.

Our new method is to start from the results of the baseline and adapt them for a bigger payload. This allows us to make no trade-offs by accommodating the 70 passengers while reusing as much hardware as possible.

4.2.2 Process

The first step is to compute the new payload mass.

Then we estimate the new operating empty mass. As the number of passengers increases, the length of the fuselage increases in the same proportions which has an influence on the operating empty mass. We will however assume a scaling factor of 0.4 [ASS-02]. This means that the increase of the mOE is only 40% of the increase of the payload mass, as the fuselage represents only a fraction of the mOE, wings and propulsion system being common between the variants.

The longer fuselage increases drag and so fuel consumption. We will correct our fuel factor accordingly. This will allow us to calculate our fuel mass. From there it is easy to get the take-off mass.

4.3 Results

4.3.1 Variant Masses

Estimated characteristic masses of the stretched variant at design point :

Parameter	**Stretched variant at design point**
Operational Range	700 NM
Physical Range	935 NM
Operating Empty Mass (m_{OE})	14,831 kg
Payload Mass (m_{PL})	7,150 kg
Fuel Mass (m_{fuel})	728 kg
Maximum Take-off Mass (m_{MTO})	22,710 kg

Table 6: Stretched Variant Mass

Payload-range diagram for the variant:

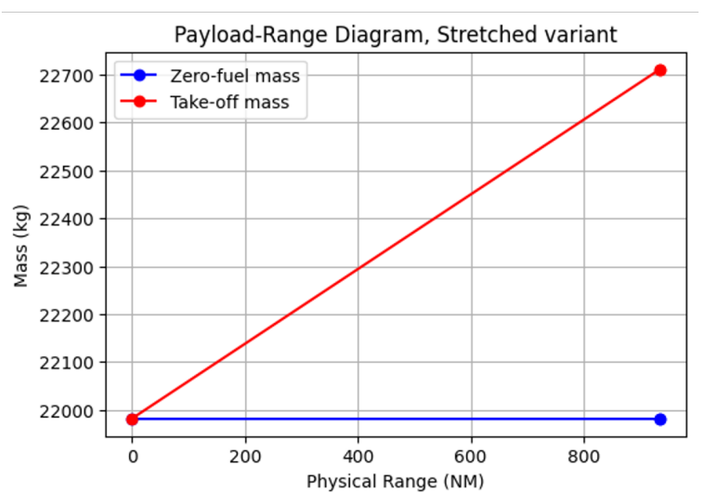

Figure 3: Stretched Variant

5 Discussion about commonalities

As for all other aforementioned masses, the fuel mass for our baseline is the minimum required to meet the 700NM range requirement [RQ-07]. However, if we choose to have the LH2 tanks as commonalities between the variants, we could trade payload mass for fuel and extend the range of the baseline. This is represented at point C in the new baseline Payload-Range diagram.

Updated Payload-range diagram for the baseline with common LH2 tanks :

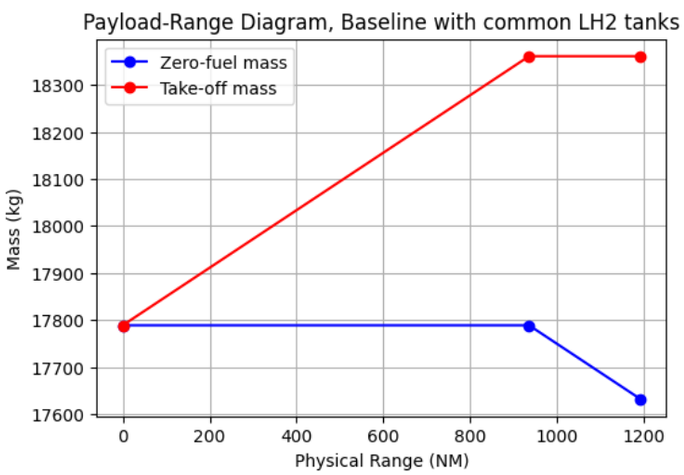

Figure 4: Baseline with common LH2 tanks

Parameter	**Baseline at point C**
Operational Range	944 NM
Physical Range	1191 NM
Operating Empty Mass (m_{OE})	12,728 kg
Payload Mass (m_{PL})	4903 kg
Fuel Mass (m_{fuel})	728 kg
Maximum Take-off Mass (m_{MTO})	18,360 kg

Table 7: Stretched Variant Mass

To compute the range in this situation, we assumed proportionality between the fuel mass and the physical range [ASS-03].

Part III
Initial Propulsive Performance Assessment

6 Methods

6.1 Estimate the power required for the aircraft

In addition to mass, the propulsion system is another important factor in the initial design. Depending on the flight phase, different levels of power are required in relation to the mass of the aircraft during take-off[9]. To obtain the required power to weight ratio of the aircraft, we will use the following methods:

- Assuming a wing loading, Reciprocal glide ratio and a flight level to get the Mach number
- Calculate the wing area
- Calculate the power to weight ratio for each flight phase

6.2 Approach speed and landing Requirements

During the landing approach, the aircraft has already used up all its fuel, which means that the aircraft's mass is now only mzF, thereby also reducing the wing loading.

6.3 Shaft Power

The calculated power-to-weight ratio can be used to calculate the total power requirement that must be provided by the two installed motors. The efficiency of the propeller and the gearbox must be taken into account here.

7 Assumptions

Baseline & Stretched Variant

- Glide ratio of $\epsilon = 18$ is used, which takes further improvements in the development in the aerodynamic into account. The certification and production of our aircraft is not expected to be finished before 2040, assuming a design freeze in 2030
- Lift coefficient of 0.5 is used.
- An Altitude of 29000ft is used which leads to a Mach number of 0.6.

- Propeller efficiency of 0.9 is used
- $\epsilon_{OEI} = 1.1\epsilon$
- The speed with OEI is reduced, which also reduces the required power-to-weight ratio.

Parameter	**Cruise**	**Cruise OEI**	**Takeoff OEI**	**Landing**
Aspect Ratio	8	8	8	8
Mass	N/A	N/A	Max Take-off Weight	Zero Fuel Mass
Epsilon (ϵ)	18	16	10	8
$C_{L,max}$	N/A	N/A	2.3	2.8
C_L	0.51	0.51	N/A	N/A
C_{D0}	0.012	0.012	N/A	N/A
Oswald Factor (e)	0.7	0.7	0.7	0.7
Altitude	29000 ft	29000 ft	0 ft	0 ft
Power throttle	0.8	0.95	1	N/A

Table 8: Baseline Assumptions

Parameter	**Cruise**	**Cruise OEI**	**Takeoff OEI**	**Landing**
Aspect Ratio	8	8	8	8
Mass	N/A	N/A	Max Take-off Weight	Zero Fuel Mass
Epsilon (ϵ)	18	16	10	8
$C_{L,max}$	N/A	N/A	2.3	2.8
C_L	0.64	0.64	N/A	N/A
C_{D0}	0.013	0.013	N/A	N/A
Oswald Factor (e)	0.7	0.7	0.7	0.7
Altitude	29000 ft	29000 ft	0 ft	0 ft
Power throttle	0.8	0.95	1	N/A

Table 9: Stretched Variant Assumptions

8 Results

8.1 Chart P/W vs W/S

Baseline A graphical method was used to obtain the required power-to-weight ratio for take-off. The two resulting lines are plotted on a diagram, and the point at which they intersect determines the power-to-weight ratio for take-off. Which leads to a needed Power to weight ratio of 19 W/N.

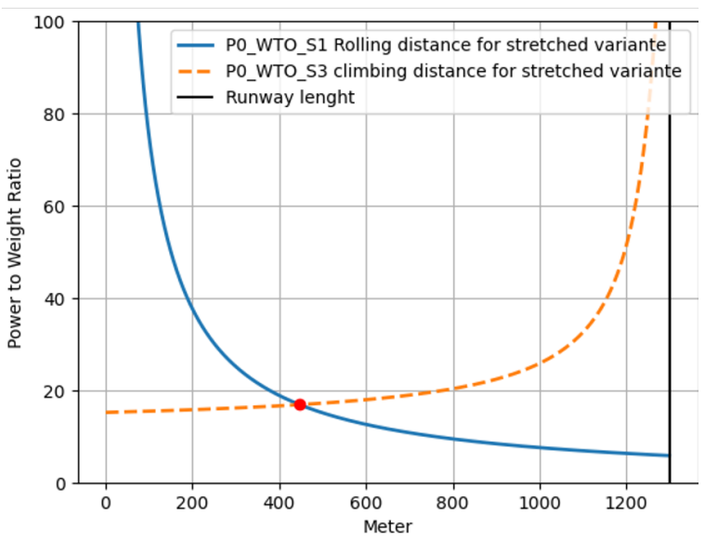

Figure 5: Baseline Power to Weight Ratio

By using a wing loading of $3,638 N/m^2$ we need a power to weight ratio of at least 23 W/N to maintain those flight conditions.

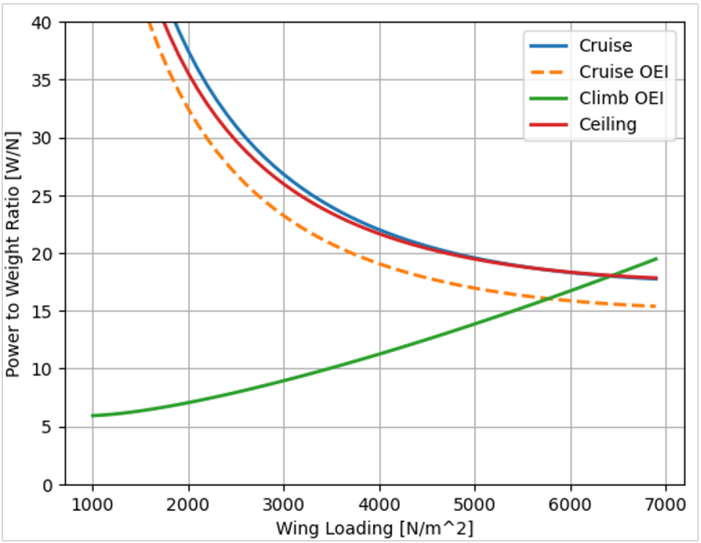

Figure 6: Baseline Wingload Power to Weight Ratio

Stretched Variant For the stretch variant the wing loading is higher due to the higher mass while using the same wing for both variants. It leads to a wing loading of $4,500 N/m^2$

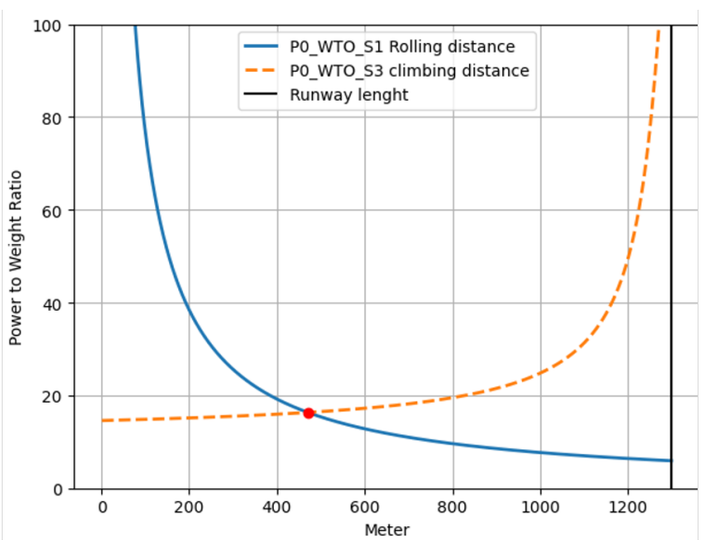

Figure 7: Stretched Variant Power to Weight Ratio

A power to weight ratio of 19 W/N is needed to take off in given runway length.

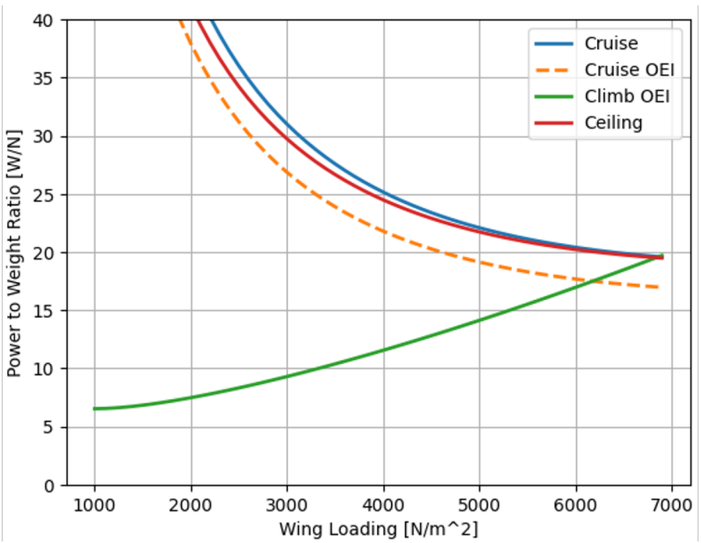

Figure 8: Stretched Variant Wingload Power to Weight Ratio

By using a wing loading of $4,500 N/m^2$ we need a power to weight ratio of at least 23W/N to maintain those flight conditions.

8.2 Approach speed and landing requirement

Baseline With a maximum lift coefficient of $C_{L,max} = 2.8$ and a wing loading of $W/S = 3400\,\mathrm{N/m^2}$. Due to the fuel consumed, the wing loading during approach and landing is reduced in accordance with the κ (fuel factor): Using the Lift Formula

$$L = \frac{1}{2}\rho v^2 S C_L \tag{Lift Formula}$$

and solving for the velocity at maximum lift (i.e., stall):

$$v_{\text{stall}} = \sqrt{\frac{2}{\rho C_{L,\max}} \frac{W}{S}} \tag{Stall Speed}$$

The approach speed v_{app} is calculated as:

$$v_{app} = 1.23 \times 1.944 \times \sqrt{\frac{2 \times 3400}{1.225 \times 2.8}} = 106\,\mathrm{kt} \tag{Baseline v_{app}}$$

This results in a landing distance of 966m.

Stretched Variant With a maximum lift coefficient of $C_{L,max} = 2.8$, a wing loading of $W/S = 4200\,\mathrm{N/m^2}$ the approach speed are 118kt

$$v_{app} = 1.23 \times 1.944 \times \sqrt{\frac{2 \times 4200}{1.225 \times 2.8}} = 118\,\mathrm{kt} \tag{Stretched v_{app}}$$

This results in a landing distance of $1,111$m.

8.3 Shaft Power

Baseline

- Transmission Efficiency: $\eta_{trans} = 0.99$
- Power-to-Weight Ratio: $P/W = 23\,\mathrm{W/N}$
- Maximum Takeoff Mass: $m_{MTO} = 18360\,\mathrm{kg}$
- Number of Engines: $N_{eng} = 2$
- The power per engine (P_{eng}) is calculated as:

$$P_{eng} = \frac{\eta_{trans} \cdot (P/W) \cdot (m_{MTO} \cdot g)}{N_{eng} \cdot 1000} \qquad \text{(Power Formula)}$$

$$P_{eng} = \frac{0.99 \cdot 23 \cdot (18360 \cdot 9.81)}{2 \cdot 1000}$$

$$P_{eng} \approx 2050\,\mathrm{kW} \qquad \text{(Baseline Power)}$$

Stretched Variant

- Transmission Efficiency: $\eta_{trans} = 0.99$
- Power-to-Weight Ratio: $P/W = 23\,\mathrm{W/N}$
- Maximum Takeoff Mass: $m_{MTO} = 22688\,\mathrm{kg}$
- Number of Engines: $N_{eng} = 2$
- The power per engine (P_{eng}) is calculated as:

$$P_{eng} = \frac{0.99 \cdot 23 \cdot (22688 \cdot 9.81)}{2 \cdot 1000} \qquad \text{(Power Formula)}$$

$$P_{eng} \approx 2536\,\mathrm{kW} \qquad \text{(Stretch Power)}$$

8.4 Propeller Diameter

The diameter of the propeller can now be calculated using the total power per motor.

Baseline The Propeller diameter D is estimated based on the power per engine P_{eng} using a propeller sizing coefficient $K_p = 0.49$:

$$D = K_p \cdot \sqrt[4]{P_{eng}} \qquad \text{(Prop. Formula)}$$

$$D = 0.49 \cdot \sqrt[4]{2050} \approx 3.29\,\mathrm{m} \qquad \text{(Baseline Prop)}$$

Stretched Variant The Propeller diameter D is estimated based on the power per engine P_{eng} using a propeller sizing coefficient $K_p = 0.49$:

$$D = K_p \cdot \sqrt[4]{P_{eng}}$$

$$D = 0.49 \cdot \sqrt[4]{2536} \approx 3.48\,\mathrm{m} \qquad \text{(Stretched Prop)}$$

Part IV
Hydrogen systems

9 HEF System design choices

9.1 Architecture layout

The electrical motors of our aircraft are powered by fuel cells running on liquid hydrogen stored in cooled and pressurized tanks. As previously stated, we want to keep these systems as commonalities. The stretched variant is the most demanding so we will use its requirements to scale the systems.

The HEF systems of our aircraft will include :

- Two separate tanks to satisfy regulation requirements (i.e. CS 25.953 regulations concerning fuel systems independence), their installation systems and the LH2 distribution systems.
- A small "booster pack" buffer battery to provide a 10% power increase during 90s (for emergencies)
- Fuel cells that will be over-sized to improve efficiency. They are composed of a stack, ancillaries and a cooling system.
- Inverters to convert the direct current of the fuel cells into alternative current to run the engines.
- Electrical motors with their gearbox assembly and cooling system.

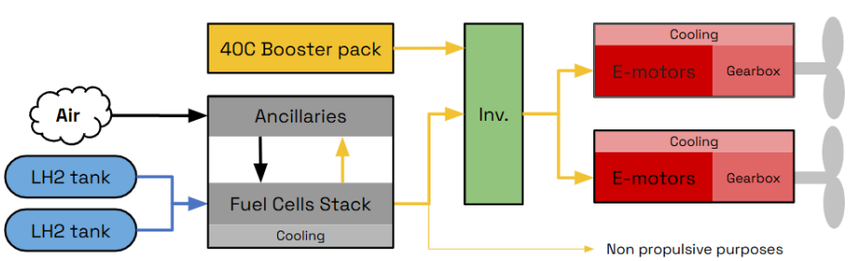

Figure 9: HEF architecture layout

The choice of a system that depends largely on fuel cells (that is to say that there is no relevant amount of energy stocked in the batteries) has been made to satisfy safety requirements while keeping the operating-empty mass of the aircraft as low as possible.

9.2 Fuel cells load optimization

The mass/volume specific powers and the efficiency of the fuel cells depend on their load : the more the fuel cells stack is oversized, the more efficient - but also the bigger and the heavier - it becomes. But the relation is even more complicated when we consider that a better efficiency decreases both the heat generated (and thus the cooling system) and the fuel consumption.

We have then created the following diagram that represents the mass and volume of the whole fuel cells system against the load, as well as the associated stack efficiency :

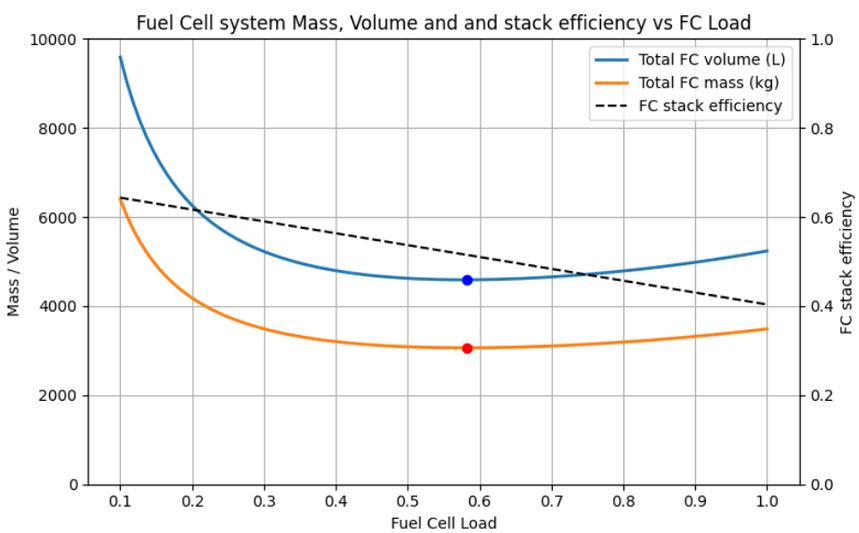

Figure 10: Fuel cells characteristics vs load

Over the possible 10% - 100% range, the minimum mass (3063 kg) and volume (4.59 m^3) for the whole system are reached with a load of 58.2%, which gives a stack efficiency $\eta_{\text{FC,stack}} = 0.515$. As the mass and volume of the system do not vary significantly between 40% and 75% load, we choose to load our fuel cells at 45% during cruise to get a better efficiency ($\eta_{\text{FC,stack}} = 0.55$).

10 HEF System Volume & Mass Estimation

10.1 Methods

In order to integrate these systems to our aircraft we need to estimate their volumes. We also conducted a first estimation of their mass as the required calculations are similar.

10.1.1 Hydrogen Tanks

The first step is to estimate the volume and mass of the Liquid Hydrogen (LH2) tank systems, which are composed of the tanks themselves, the installation hardware, and the fuel distribution systems.

- **Tank Volume (V_{tank}):** The internal volume must be larger than the fuel volume to account for the gas expansion space. With a volumetric ratio of 1.072:

$$V_{\text{tank}} = 1.072 \cdot \frac{m_F}{\rho_{\text{LH2}}} \tag{Tank Volume}$$

- **Tank Mass (m_{tank}):** The LH2 tanks are assumed to have a Gravimetric Efficiency (η_{grav}) of 0.5 [ASS-08]. This allows us to calculate the dry mass of the tanks based on the fuel mass (m_F):

$$m_{\text{tank}} = m_F \cdot \left(\frac{1}{\eta_{\text{grav}}} - 1 \right) \tag{Tank Mass}$$

- **System Masses:** The installation systems and fuel distribution systems are estimated as fractions of the tank mass:

$$m_{\text{install}} = 0.05 \cdot m_{\text{tank}} \tag{Install Mass}$$

$$m_{\text{dist}} = 0.15 \cdot m_{\text{tank}} \tag{Dist. Mass}$$

10.1.2 Fuel Cells

To conduct the sizing calculations, we need to trace the power requirements upstream from the mechanical output. Starting with the previously determined motor output power ($P_{\text{mech,mot}}$), the power chain is calculated as follows:

- **Motor Input Power:** Motor efficiency ($\eta_{\text{mot}} = 0.99$):

$$P_{\text{el,mot}} = \frac{P_{\text{mech,mot}}}{\eta_{\text{mot}}} = \frac{P_{\text{mech,mot}}}{0.99} \tag{Motor Input}$$

- **Inverter Input Power:** With the inverter efficiency ($\eta_{\text{inv}} = 0.99$), we get the power consumption from the Fuel Cells for propulsion (= the inverter input power):

$$P_{\text{el,inv}} = \frac{P_{\text{el,mot}}}{\eta_{\text{inv}}} = \frac{P_{\text{el,mot}}}{0.99} \tag{Inv. Input}$$

- **Total Fuel Cell System Power:** The Fuel Cells supply both the propulsive power and the non-propulsive loads. The non-propulsive load is estimated at 6.25% of the inverter input.
  The total power requirement depends on the Throttle Ratio (TR), assumed as:
  - 0.8 (Cruise)
  - 0.9 (Climb)
  - 1.0 (Takeoff/emergency)

  $$P_{\text{el,fc,syst}} = P_{\text{el,inv}} \cdot (TR + 0.0625) \qquad \text{(Total FC Power)}$$

  With the specific power density of the ancillaries at the Climb throttle ratio, we can directly size the mass and volume of the ancillaries.

- **Fuel Cell Stack Power:** The stack sizing is driven by the power output during Cruise and the ancillary efficiency factor ($\eta_{\text{FC,anc}} = 0.93$) [ASS-12]:

  $$P_{\text{el,stack}} = \frac{P^{\text{cruise}}_{\text{el,fc,syst}}}{\eta_{\text{FC,anc}}} = \frac{P^{\text{cruise}}_{\text{el,fc,syst}}}{0.93} \qquad \text{(Stack Power)}$$

  We use the specific powers to calculate the mass and the volume from the stack.

- **Cooling System Sizing:** The cooling system is sized to handle 80% of the heat flow generated at Full Throttle. The heat flow $\dot{Q}$ is derived from the inefficiency of the system:

  $$\dot{Q} = 0.8 \cdot P_{\text{el,inv}} \cdot (1 + 0.0625) \cdot \left(\frac{1}{a \cdot FC_{\text{load}} \cdot \frac{1}{0.8} + b} - 1 \right) \qquad \text{(Heat Flow)}$$

  We use the specific powers of the heating system to calculate its mass and volume.

10.1.3 Batteries

We choose a booster pack providing a 10% increase in power (that is to say $512kW$). From this we use the specific powers of the booster batteries to calculate the mass and the volume.

10.1.4 Inverters

In the same manner that we did for the booster pack, we use the specific powers of the inverters to calculate their mass and volume based on the power input they receive.

10.1.5 Electric Motors

We again use the same method for the electric motors + gearbox assembly. The cooling system adds another 20% of the mass.

10.2 Results

HEF Systems	Design Reference	Mass (kg)	Volume (m^3)
LH2 Tanks	Fuel mass	728	11.035
LH2 Tank Installation	5% Tanks mass	36	N/A
LH2 Distribution	15% Tanks mass	109	N/A
Fuel Cell Stack	FC System output (Cruise)	1095	1.643
Fuel Cell Ancillaries	FC Stack output (Climb)	846	1.245
Fuel Cell Cooling	80% Heat flow at Max FC System output	1186	1.798
Fuel Cells Total	N/A	**3129**	**4.687**
Booster Pack	10% E-Motors power output	51	0.051
Inverters	Inverters electric input	207	0.207
E-Motors	E-Motors electric input	409	0.204
E-Motors Cooling	20% E-Motors mass	123	N/A

Table 10: Mass and Volume Breakdown of HEF Systems

Part V
Configuration

11 Fuselage

11.1 Methods & Assumptions

Assumptions

Cabin Interior Specifications

- **Seat Pitch:** 36 inches.
  - *Rationale:* Selected as the higher end of typical economy class to ensure passenger comfort on short-range routes as there is no additional business class needed [RQ-34][RQ-68].
- **Seating Configuration:** 4 Seats per row (2 × 2 Layout).
  - *Rationale:* Maximizes seat width (W_{seat}) based on the fuselage inner diameter (D_{in}) and standardized aisle width (W_{aisle}):

$$W_{\text{seat}} = \frac{D_{\text{in}} - W_{\text{aisle}}}{4} \qquad \text{(Seat Width)}$$

- **Toilets:** 2.

- *Rationale:* The design with two toilets makes sense, as it reduces waiting times and increases passenger comfort. In addition, a second toilet improves system redundancy and ensures basic sanitary facilities remain available in the event of a failure.

- **Cabin Crew:** 2 Flight Attendants.
  - *Rationale:* While the baseline capacity would be well served and compliant with regulations [RegRQ-5] by having a single attendant, two are selected to ensure redundancy, service quality, and compliance with the Stretched Variant.

- **Emergency Exits:** 2 Pairs of Type I doors.
  - *Rationale* Compliant with [RegRQ-06] and [RegRQ-07] . From [RQ-24] to [RQ-28] for baseline and from [RQ-58] to [RQ-62] for stretch variant.

11.2 Results

Based on a configuration of 4 seats per row with a 36-inch pitch, the **Base Variant** (48 passengers) requires a cabin length $l_{cabin,baseline}$ of 14 m, while the **Stretch Variant** $l_{cabin,stretch}$ requires 22 m.

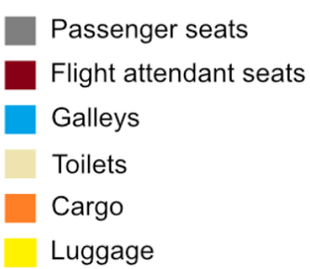

Figure 11: Label

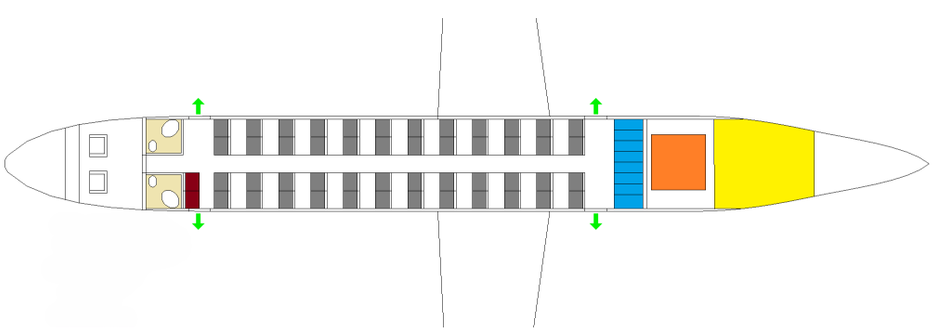

Figure 12: Sketch: Baseline Plane, Cabin layout

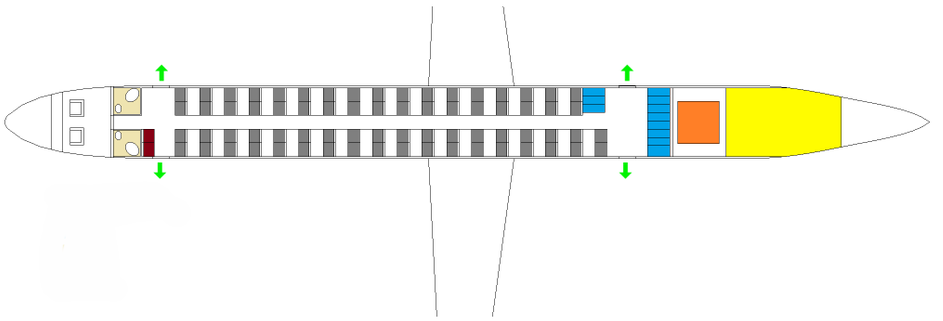

Figure 13: Sketch: Stretch Plane, Cabin layout

The fuselage dimensions are calculated as follows:

- **Fuselage Diameter (D):** Derived from the inner diameter (d_{in}) and wall thickness (t_w):
$$D = d_{\text{in}} + 2 \cdot t_w = 2.54 + 2 \cdot 0.076 = 2.69\,\text{m} \qquad \text{(Fuselage Dia.)}$$

- **Aerodynamic Section Lengths:** The nose (i.e., cockpit) and tail sections are sized proportionally to the diameter:
$$l_{\text{cockpit}} = 1.5 \cdot D = 1.5 \cdot 2.69 \approx 4.03\,\text{m} \qquad \text{(Cockpit Len.)}$$
$$l_{\text{tail}} = 3 \cdot D = 3 \cdot 2.69 = 8.07\,\text{m} \qquad \text{(Tail Len.)}$$

- **Total Aircraft Length (L_{total}):** Summing the sections ($L_{\text{total}} = l_{\text{cockpit}} + l_{\text{cabin}} + l_{\text{tail}}$):
$$L_{\text{total,Base}} = 4.03 + 14.00 + 8.07 = 26.10\,\text{m} \qquad \text{(Base Length)}$$
$$L_{\text{total,Stretch}} = 4.03 + 22.00 + 8.07 = 34.10\,\text{m} \qquad \text{(Stretch Length)}$$

- **Tail clearance:** To prevent tail strikes, our aircraft has a clearance angle of 13 degrees. The landing gear placement is likely to change, but we have margins.

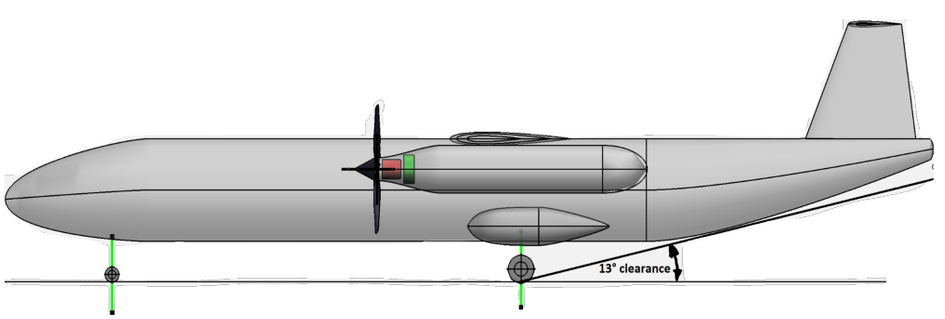

Figure 14: Side-view

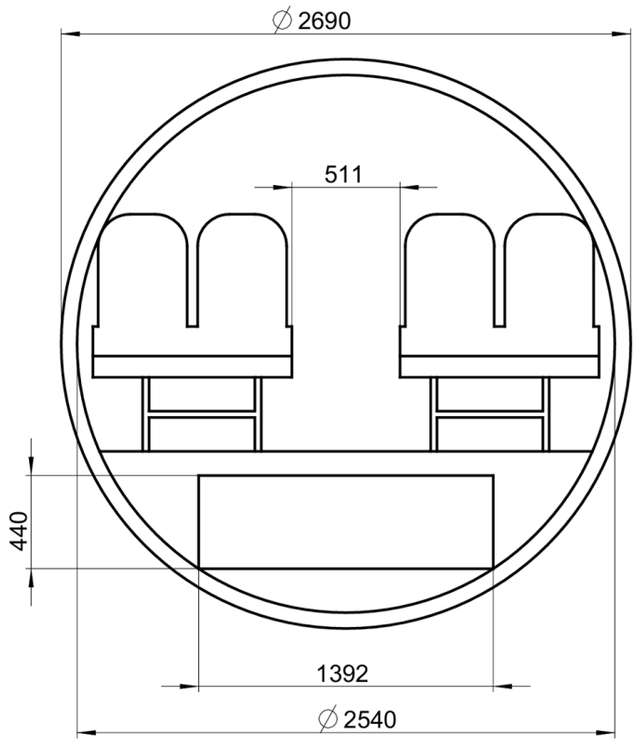

Figure 15: Cross section

Due to the four seats per row, the fuselage diameter is comparatively small. This means that the available storage space in the belly is very limited and is partly occupied by the hydrogen system. We have decided to transport luggage and cargo in the cabin, which has been slightly extended for this purpose. A total of $500kg$ of cargo in a volume of $3.1m^3$ and $576kg$ of luggage in a volume of $3.49m^3$ must be transported for the base variant, and $840kg$ in $5.09m^3$ for the stretch variant. The cargo could be transported on a PEx pallet that fits perfectly in the rear of the fuselage.

12 Wing

12.1 Methods

We chose a basic profile such as NACA 43013. NACA 43 series profiles are used for the ATR42-320 or 600 aircraft. Perhaps we should choose different profiles at the root and at the tip wing, but this requires an aerodynamic study that we have not yet carried out.

12.1.1 Lift Coefficient

$\mathbf{C_{l,ICA}}$ Value
First, we determine the lift coefficient (C_L) in cruise flight using the lift equation:

$$C_L = \frac{2}{\gamma \cdot P \cdot M^2} \left(\frac{W}{S}\right) \qquad \text{(Cruise } C_L\text{)}$$

- $\gamma = 1.4$: The ratio of specific heats (adiabatic index) for dry air, appropriate for subsonic and transonic flight regimes.
- $P = 30\,700\,\text{Pa}$ (30.7 kPa): The static pressure at the cruise altitude of 29 000 ft (FL290).
- $M = M_{MO} = 0.57$: was selected higher than the cruise mach number used for the initial propulsion system sizing in order to provide an operational margin.
- W/S: The wing loading was selected based on the Power Loading P/W vs. Wing Loading W/S) diagram in section 8.1
  - The intersection of the Service Ceiling and One Engine Inoperative (OEI) Climb constraints occurs at wing loadings of: $\approx 6100\,\text{N/m}^2$ for the Baseline and $\approx 6500\,\text{N/m}^2$ for the Stretch variant.
    However, such values would be unrealistic for a regional turboprop aircraft due to excessive take-off and landing speeds.
  - To maintain a single wing design for both aircraft variants (minimizing complexity), a value of $W/S = 4500\,\text{N/m}^2$ for the stretched variant (which correspond to $W/S = 3683\,\text{N/m}^2$ for the baseline) has been chosen as it satisfies all performance requirements of the stretch variant while reducing the required power-to-weight ratio. The values therefore represents a suitable compromise for the aircraft family.

$\mathbf{C_{l,BO}}$ Value

The buffet onset lift coefficient was determined by applying safety factors to the cruise lift coefficient to account for operational and certification margins.
We use the following equation:

$$\mathbf{C}_{\text{l,BO}} = k_n \cdot k_{\text{AV}} \cdot k_{\text{trim}} \cdot C_L \qquad \text{(Buffet Onset)}$$

- C_L: The lift coefficient in cruise conditions.
- k_n: Factor accounting for load factor variations due to maneuvers and vertical gusts.
- k_{AV}: Factor representing aerodynamic uncertainties and modeling margins.
- k_{trim}: Factor accounting for the additional lift required to balance the aircraft (trim drag/lift effects).

$\mathbf{C_{l,DD}}$ Value

The drag divergence lift coefficient was evaluated to verify whether drag divergence is significantly exceeded during cruise flight. We use the following equation:

$$\mathbf{C}_{\mathrm{l,DD}} = k_{\mathrm{AV}} \cdot k_{\mathrm{trim}} \cdot C_L \tag{Drag Div.}$$

With k_{AV} and k_{trim} are identical to the factors applied in the buffet onset condition.

As a local increase in drag is acceptable during high-speed cruise, drag divergence is not considered a sizing criterion for the wing sweep. The sweep angle is instead determined by the buffet onset condition.

12.1.2 Sweep

Profile Mach Number The NACA 43013 profile mach number was determined based on the buffet onset lift coefficient. Using a buffet boundary diagram representing the lift coefficient as a function of Mach number. We can read the profile Mach number corresponding to $\mathbf{C_{l,BO}}$.

Calculation of Sweep The wing sweep angle was evaluated based on the profile Mach number (M_n) obtained from the buffet onset condition:

$$\cos(\phi_{25}) = \frac{M_n}{M_{\mathrm{MO}}} \tag{Sweep Angle}$$

Since the calculated profile Mach number M_n is higher than the maximum operating Mach number M_{MO}, the ratio exceeds 1, implying that **no sweep is required** to reduce the normal Mach number.

12.1.3 Geometry

Aspect Ratio The wing aspect ratio was assumed equal to $AR = 8$ during the initial propulsion system sizing [ASS-06]. This value lies within the typical range recommended for regional aircraft and is consistent with the guidelines provided in the course. An aspect ratio of 8 represents a suitable compromise between induced drag reduction and structural weight, and is therefore appropriate for the considered aircraft class.

Taper Ratio The wing taper ratio was selected equal to that of the ATR-42 reference aircraft $\lambda = 0.45$ [ASS-07]. This value is representative of regional aircraft and provides a suitable compromise between aerodynamic efficiency, structural weight and stall behavior.

Calculation of Geometry Parameters Since we have defined the wing loading (W/S), we can determine the wing reference area (S), span (b), and chord dimensions using the Aspect Ratio (AR) and Taper Ratio (λ):

$$S = \frac{W}{(W/S)} \tag{Ref. Area}$$

$$b = \sqrt{AR \cdot S} \tag{Span}$$

$$c_{\text{mid}} = \frac{S}{b} \tag{Mean Chord}$$

$$c_{\text{root}} = \frac{2 \cdot S}{b \cdot (\lambda + 1)} \tag{Root Chord}$$

$$c_{\text{tip}} = \lambda \cdot c_{\text{root}} \tag{Tip Chord}$$

12.2 Results

12.2.1 Lift coefficients

Parameter	Baseline Variant	Stretched variant
Lift coefficient C_L	0.52	0.64
Drag Divergence lift coefficient $C_{l,DD}$	0.63	0.78
Buffet Onset lift coefficient $C_{l,BO}$	0.82	1.014

Table 11: Lift coefficients

12.2.2 Geometry

Parameter	Value
Wing area S	49.46 m^2
Wing span b	19.89 m
Mean chord C_{mid}	2.49 m
Root chord C_{root}	3.43 m
Tip chord C_{tip}	1.54 m

Table 12: Wings dimensions

Based on these values, we can sketch the wing geometry. Since the decision was made to attach the engines to the wings, a **double trapezoidal** planform was chosen (to optimize local aerodynamics at the engine mounting point).

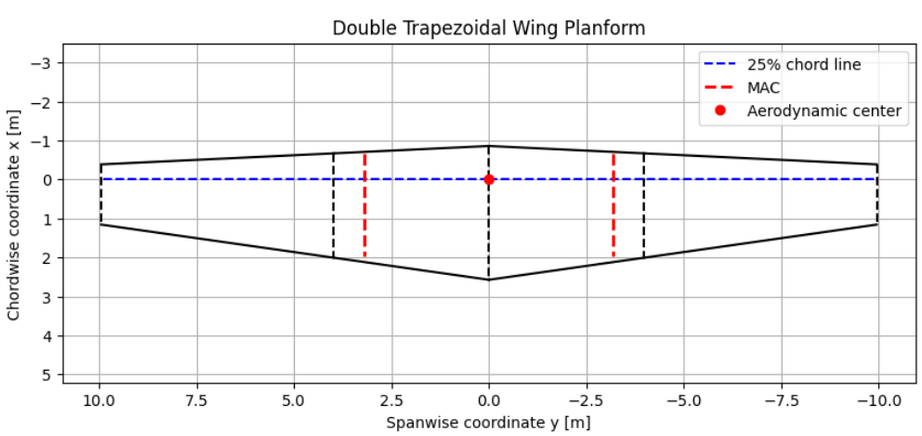

Figure 16: Wing Form

The "kink" represented by the black line indicates where the engines are positioned. The angle at this location is so small that we could simplify the double trapezoidal geometry into a single wing panel. This would simplifies the manufacturing process.

13 Empennage

13.1 Methods

The Empennage is important for the stability and controllability of the aircraft. It consists of the horizontal and vertical tailplane. The horizontal tailplane is responsible for the pitch of the aircraft, while the vertical tailplane is responsible for yaw.

The size of the horizontal tailplane is calculated from the mean chord of the wing and its area, the distance between the aerodynamic centers and the volume coefficients, which can be obtained from tables :

$$S_h = \frac{c_h \cdot S_w \cdot c_{\text{mid}}}{l_h} \tag{HTP Area}$$

$$S_v = \frac{c_v \cdot S_w \cdot b}{l_v} \tag{VTP Area}$$

13.2 Results

Parameter	**Value**
Horizontal tailplane area Sh	9.84 m^2
Vertical tailplane area Sv	6.89 m

Table 13: Empennage dimensions

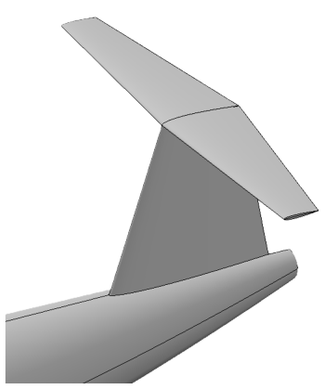

Figure 17: Empennage

14 HEF System integration

Now that we know the aircraft's dimensions and the volumes necessary to accommodate the HEF systems, we can think about their integration in the aircraft :

The engines are located under the wings. We can take advantage of the pods that are necessary to accommodate them : the pods could include the inverters whose volume is relatively small. We could also elongate the pods to fit the LH2 tanks. Thus, each pod would contain $0.1m^3$ of engine + $0.1m^3$ of inverter + a tank of $5.5m^3$. The figure below shows a pod of 1.25m in diameter and 7.55m long. While increasing the drag, this pod configuration has the advantage of leaving room in the fuselage for the luggage/freight, it is also scalable if needed (it is easier to change the length of the pod than to redesign the whole aircraft). Finally, it avoids placing the cooled LH2 tanks close to the cabin whose temperature is regulated.

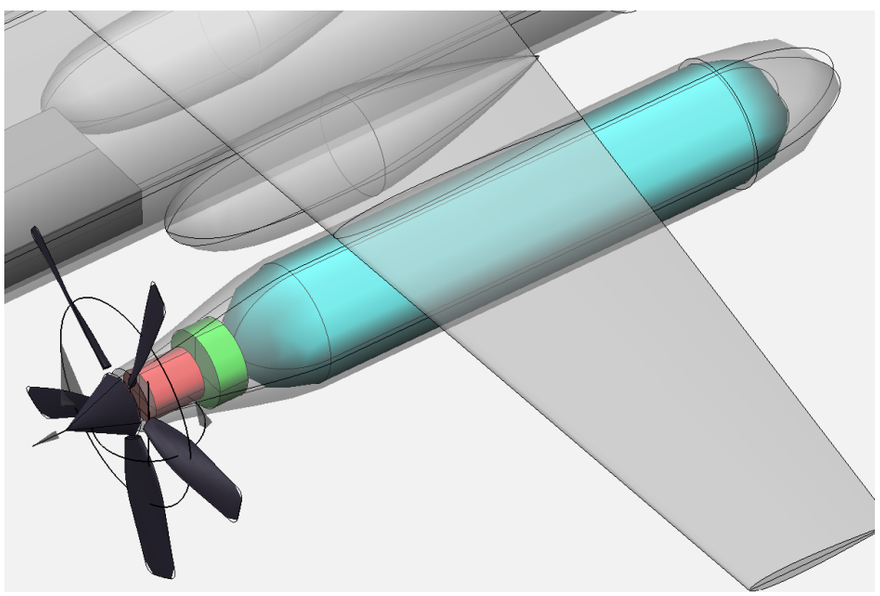

Figure 18: Engine pod design

The booster pack is very small and can be fitted freely in the aircraft. We choose to put it in the belly.

Fuel cells could also be fitted in the nearly empty belly of the plane. This would help lower the center of mass of the plane to increase stability. There are two solutions : closer to the wings in order to keep the fuel distribution pipes small, or in the front to balance the weight of the cargo. In any case, we should pay attention to leaving room for the landing gears.

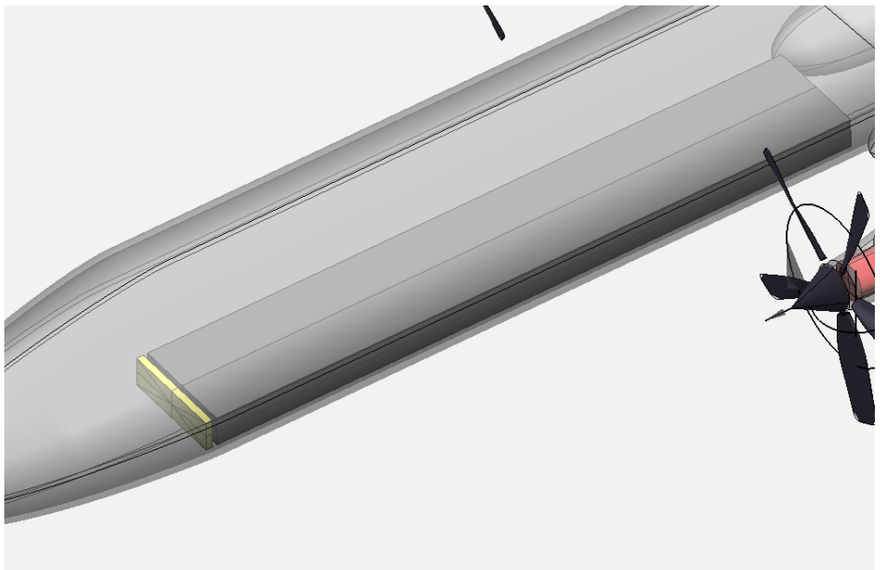

Figure 19: Fuel cells and battery integration

Part VI
Conclusion

15 Design Summary

In summary, the final design parameters of both the baseline and stretched aircraft variant are presented in the following Table 14. It displays the key specifications, taking into account the commonalities. Effective values taking into account the commonalities:

Parameter	**Baseline Variant**	**Stretch Variant**
Operation conditions		
Operational range	700 NM	700 NM
Passenger number	48	70
Freight mass	500 kg	500 kg
Cruise speed	Mach 0.515	Mach 0.515
Cruise altitude	29,000 ft	29,000 ft
Masses		
Maximum take-off mass m_{MTO}	18,360 kg	22,710 kg
Operating empty mass m_{OE}	12,728 kg	14,831 kg
Payload mass m_{PL} (at point B)	5,060 kg	7,150 kg
Fuel mass m_{fuel} (at point B)	572 kg	728 kg
Propulsion		
Number of engines	2	2
Power per engine P_{eng}	2536 kW	2536 kW
Power-to-weight ratio P/W	28 W/N	23 W/N
Propeller diameter D_{prop}	3.48 m	3.48 m
HEF systems		
Booster pack power	507.2 kW	507.2 kW
Fuel cells load	45 %	45 %
Aerodynamics		
Lift coefficient C_L	0.52	0.64
Aspect ratio AR	8	8
Taper ratio λ	0.45	0.45
Sweep at 25% chord line Λ	0°	0°
Wing area S	49.46 m²	49.46 m²
Wing loading W/S	3638 N/m²	4500 N/m²
Horizontal tailplane area Sh	9.84 m²	9.84 m²
Vertical tailplane area Sv	6.89 m²	6.89 m²
General dimensions		
Length	26.10 m	34.10 m
Width (wingspan)	19.90 m	19.90 m
Outer fuselage diameter	2.69 m	2.69 m
Inner fuselage diameter	2.54 m	2.54 m

Table 14: Final key parameters for the baseline and stretch variants

16 Requirement Compliance

Both the baseline and stretch variants meet all the specified design requirements. The baseline variant carries 48 passengers plus 500kg of freight over an operational range of 700NM (1300km) with standard reserves, cruising above Mach 0.5 at 25,000ft, as required.

The stretch variant is capable of accommodating 70 passengers (with equivalent freight) and achieving a similar range performance.
Each aircraft complies with the relevant regulatory standards, including EASA CS-25 [5] and the noise requirements of CS-36[2] and ICAO Annex 16[10].
Key performance constraints—such as take-off distance, landing distance, climb gradient (including one-engine inoperative), and cruise ceiling—are satisfied by both designs.
For instance, approach speeds for both variants (approximately 106kt for the baseline and 118kt for the stretch) keep landing distances below the 1300m requirement, and the power-to-weight ratio (23 W/N) ensures compliance with all climb and cruise performance targets.

17 Family Concept Assessment

An Objective & Key-Result (OKR) of the study was to develop a family of aircrafts that ideally maximizes commonality between the baseline and stretch versions.
The stretch variant is essentially a fuselage extension of the baseline design, reusing the same wing planform, propulsion architecture, and hydrogen fuel system.
Also the wing and control surfaces remain the same. The liquid hydrogen tank design is also similar; the stretch variant can accommodate the extra fuel needed for its mission by leveraging the same tank without fundamentally changing the storage system.
This reuse of systems for both variants offers significant advantages: it reduces development and manufacturing costs, shortens the certification timeline and provides more options in the daily operations. Airlines operating both variants would benefit from common pilot type ratings and maintenance procedures.

References

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[2] EASA, *CS-36, Amendment 6*. Cologne, Germany: European Union Aviation Safety Agency, 2021. [Online]. Available: https://www.easa.europa.eu/en/document-library/certification-specifications/cs-36-amendment-6.

[3] M. Berens, *Lecture Notes: Aircraft Database 2025W*. 2025.

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Part VII
Appendix A

18 Used Python Code

18.1 First mass estimation - Values at point B

```
# Imports
from math import *
from matplotlib import pyplot as plt
import numpy as np

# First Mass Estimation (Benoit Version)
##2025/11/21 - Initial mass estimation v2 - Benoit Pereira

from math import *
import matplotlib.pyplot as plt
import numpy as np

#Design specifications
NbPass_bas = 48                                                    #Passenger number for baseline
NbPass_str = 70                                                    #Passenger number for streched variant
mPass = 83                                                         #Passenger mass (kg)
mLugg = 12                                                         #Luggage mass per passenger (kg)
mFrght = 500                                                       #Freight mass (kg)

rOP = 700                                                          #Operational range (NM)
rDIV = 100                                                         #Diversion range (NM)
CTGratio = 0.05                                                    #Contingency ratio                    (proportion of operational range)
vHLD = 200                                                         #Holding speed (knots)
tHLD = 0.5                                                         #Holding time (hours)



#Estimations/Assumptions/Design choices
```

```
##Values from statistics
y = 0.28                                                    #Payload factor (gamma)                  (proportion of payload relative to take-off mass)
k = 0.12                                                    #Fuel factor (kappa)                     (proportion of fuel relative to take-off mass)

##Correction for better technologies
y = y*1.1                                                   #Estimation of 10% increase of the payload factor by 2030
k = k*0.8                                                   #Estimation of 20% decrease of the fuel factor by 2030



#First mass estimation
rCTG = CTGratio*rOP                                         #Contingency range
rHLD = vHLD*tHLD                                            #Holding range
rPH = rOP + rHLD + rDIV + rCTG                              #Physical range                          (includes reserves)

##Baseline
mP_bas = NbPass_bas*(mPass+mLugg) + mFrght                  #Payload mass for baseline               (passenger + baggages + freight)
mTO_bas = mP_bas/y                                          #Take-off mass
mF_bas = k*mTO_bas                                          #Fuel mass
mOE_bas = (mTO_bas-mP_bas-mF_bas)                           #Operating-empty mass

k_bas = k*1.07*1.02*0.833*0.357                             #Correction for LHE propulsion
mOE_bas = 1.3 * mOE_bas                                     #Correction for LHE propulsion
mTO_bas = (mOE_bas+mP_bas)/(1-k_bas)                        #Correction for LHE propulsion
mF_bas = k_bas*mTO_bas                                      #Need to be computed again for new LHE fuel factor

##Stretched variant
k_str = 1.03*k_bas                                          #Corrected fuel factor for stretched version

```

```
mP_str = NbPass_str*(mPass+mLugg) + mFrght                #Payload mass for stretched variant
mOE_str = mOE_bas*(1+0.4*(mP_str-mP_bas)/mP_bas)          #Operating-empty mass for stretched variant
mTO_str = (mOE_str+mP_str)/(1-k_str)                      #Take_off mass for stretched variant
mF_str = k_str*mTO_str                                    #Fuel mass for stretched variant



#Payload-Range Diagrams
##Baseline
r_vals = [0, rPH, rPH]
mZF_vals = [mP_bas+mOE_bas, mP_bas+mOE_bas, mTO_bas-mF_bas]
mTO_vals = [mP_bas+mOE_bas, mTO_bas, mTO_bas]

plt.figure(figsize=(6,4))
plt.plot(r_vals, mZF_vals, 'o-', color = 'blue', label = 'Zero-fuel mass')
plt.plot(r_vals, mTO_vals, 'o-', color = 'red', label = 'Take-off mass')
plt.xlabel("Physical Range (NM)")
plt.ylabel("Mass (kg)")
plt.title(f"Payload-Range Diagram, Baseline")
plt.legend()
plt.grid()
plt.show()

##Stretched variant
r_vals = [0, rPH, rPH]
mZF_str_vals = [mP_str+mOE_str ,mP_str+mOE_str, mTO_str-mF_str]
mTO_str_vals = [mP_str+mOE_str, mTO_str, mTO_str]

plt.figure(figsize=(6,4))
plt.plot(r_vals, mZF_str_vals, 'o-', color = 'blue', label = 'Zero-fuel mass')
plt.plot(r_vals, mTO_str_vals, 'o-', color = 'red', label = 'Take-off mass')
plt.xlabel("Physical Range (NM)")
plt.ylabel("Mass (kg)")
plt.title(f"Payload-Range Diagram, Stretched variant")
plt.legend()
plt.grid()
plt.show()
```

```python
#Masses build-up
categories = ['Baseline', 'Stretched']
mOE = np.array([mOE_bas, mOE_str])
mP = np.array([mP_bas, mP_str])
mF = np.array([mF_bas, mF_str])

plt.bar(categories, mOE, color='y', label='Operating-empty mass')
plt.bar(categories, mP, bottom=mOE, color='b', label='Payload mass')
plt.bar(categories, mF, bottom=mOE + mP, color='c', label='Fuel mass
    ')
plt.xlabel("Variant")
plt.ylabel("Mass (kg)")
plt.title("Mass build-up at design point (B)")
plt.legend()
plt.show()

#Results
print(" === Ranges === ")
print("Operational range =", int(rOP), "NM")
print("Physical range =", int(rPH), "NM")
print()
print(" === Baseline (at design point) === ")
print("Payload mass =", int(mP_bas), "kg")
print("Fuel mass =", int(mF_bas), "kg")
print("Operating-empty mass =", int(mOE_bas), "kg")
print("Take-off mass =", int(mTO_bas), "kg")
print()
print(" === Stretched variant (at design point) === ")
print("Payload mass =", int(mP_str), "kg")
print("Fuel mass =", int(mF_str), "kg",)
print("Operating-empty mass =", int(mOE_str), "kg")
print("Take-off mass =", int(mTO_str), "kg")
```

18.2 First mass estimation - Tanks commonality

```
# Discussion about LH2 tanks commonality
##2026/01/11 - Impacts of LH2 tanks commonality - Benoit Pereira

mF_max = mF_str                                          #fuel mass is now the same as for the stretched variant



#Impacts
rPH_max = mF_max/mF_bas * rPH                            #Physical range is extended (assuming proportionality between fuel mass and physical range)
rOP_max = (rPH_max - rHLD - rDIV)/(1+CTGratio)           #Operational range is extended (deducing reserves from the extended physical range)



#Payload-Range Diagram
##Baseline with common LH2 tanks
r_vals = [0, rPH, rPH_max]
mZF_vals = [mP_bas+mOE_bas ,mP_bas+mOE_bas, mTO_bas-mF_max]
mTO_vals = [mP_bas+mOE_bas, mTO_bas, mTO_bas]

plt.figure(figsize=(6,4))
plt.plot(r_vals, mZF_vals, 'o-', color = 'blue', label = 'Zero-fuel mass')
plt.plot(r_vals, mTO_vals, 'o-', color = 'red', label = 'Take-off mass')
plt.xlabel("Physical Range (NM)")
plt.ylabel("Mass (kg)")
plt.title(f"Payload-Range Diagram, Baseline with common LH2 tanks")
plt.legend()
plt.grid()
plt.show()



#Results
print(" === Baseline (at point C) === ")
print("Operational range =", int(rOP_max), "NM")
print("Physical range =", int(rPH_max), "NM")
print()
print("Payload mass =", int(mTO_bas-mOE_bas-mF_max), "kg")
print("Fuel mass =", int(mF_max), "kg")
print("Operating-empty mass =", int(mOE_bas), "kg")
```

```
print("Take-off mass =", int(mTO_bas), "kg")
```

18.3 Initial propulsion system sizing - Wing Loading - Baseline

```
# Baseline propulsion sizing
##Propulsion Sizing -Base Variante - Michael Riedl

#Assumptions
S=50                            #Wing Area [m^2]
Epsilon=1/18                    #Based on slide 22
FL=290
c=304.484                       #Speed of sound at this flight level according to Isa Calculator
Ma=0.515                        #Based on Slide 15 with FL290 and 3000N/m^2 wing loading
g=9.81                          #gravitation
n_E=2                           #Number of engines
rho_OEI=0.475448                #At 29 000ft
cL_maxTO=2.3                    #Based on slide 40
rho_0=1.22500                   #air density at see level

#given Values
hscr=35                         #obstacle hight of 35ft, based on slide 43
E_ToOEI =1/10
dh_ds=0.024                     # Based on slide 47
E_2OEI=0.8*E_ToOEI
kc=0.99                         #correction for weight loss
TR_THR=0.8                      #Power throttle
eta_Trans=0.99                  #Transmission efficency
eta_Prop=0.9                    #Propeller efficency

#Calculations
W_S=mTO_bas*g/S                 #Wing Loading base variante [N/m^2]
v=c*Ma*1.944                    #Speed in knots (m/s->knots) [knots]
v_ms=v/1.944                    #Speed in m/s
P0_WTO=(kc*Epsilon*v)/(eta_Prop*eta_Trans*TR_THR)       #Power to Weight Ratio [W/N]
P0=P0_WTO*(mOE_bas*g)
#One engine inoperative
q_OEI=rho_OEI*(v/1.944)**2/2
Epsilon_OEI=1.1*Epsilon
v_OEI=(2*q_OEI/rho_OEI)**0.5
TR_THR_OEI=0.95
P0_OEI=(n_E-1)/n_E*P0
P0_WTO_OEI=(Epsilon_OEI*kc*v_OEI*n_E)/(TR_THR_OEI*eta_Trans*eta_Prop*(n_E-1))

#Speeds
v_SR=sqrt((2*1)/(cL_maxTO*rho_0)*(mOE_bas*g)/S)
```

```
v_LOF=1.08*v_SR
v2_min=1.13*v_SR
s1=np.arange(1,1300)
s3=1300-s1
hscr_over_s3 = hscr / s3
T_WTO_s1=1.15/(g*rho_0*cL_maxTO*s1*(1-1/(2*n_E)))*(mOE_bas*g)/(S*(1-0.03-0.13))
P0_WTO_S1=(T_WTO_s1*0.71*v_LOF)/(1*0.99*0.7)
T_WTO_s3 = (np.sin(np.arctan(hscr_over_s3)) + E_ToOEI) / (1 - 1/n_E)
P0_WTO_S3=(T_WTO_s3*1/2*(v_LOF+v2_min))/(1*eta_Prop*eta_Prop)
P0_WTO_2OEI=(dh_ds+E_2OEI)*v2_min/(1*eta_Prop*eta_Trans)*n_E/(n_E-1)
P0_WTO_Ceiling=((300/(v*196.85))+Epsilon)*v/(0.9*0.99*0.9)                    #300ft/min climb speed und horizontal speed Ma=0.515 but also in feet/min

#Result
print("Propulsion system sizing for base variante")
print("Power to weight Ratio =", round(float(P0_WTO),2), "[W/N]")
print("Power to weight Ratio OEI =", round(float(P0_WTO_OEI),2), "[W/N]")
print("Power to weight Ratio 2,OEI at Climb =", round(float(P0_WTO_2OEI),2), "[W/N]")
print("Power to weight Ratio at ceiling =", round(float(P0_WTO_Ceiling),2), "[W/N]")
plt.xlabel("Meter")
plt.ylabel("Power to Weight Ratio")
plt.grid(True)

plt.plot(s1, P0_WTO_S1, label="P0_WTO_S1 Rolling distance", linewidth=2)
plt.plot(s1, P0_WTO_S3, '--', label="P0_WTO_S3 climbing distance", linewidth=2)
plt.axvline(1300,color='black',label='Runway lenght')
plt.legend()
plt.tight_layout()
plt.ylim(0, 100)
idx = np.argmin(np.abs(P0_WTO_S1 - P0_WTO_S3))   # Find the crossing point of the two curves
x_intersect = s1[idx]
y_intersect = P0_WTO_S1[idx]
print("Required Power to Weight Ratio to take of in given Runway lenght:", round(y_intersect,2),"[W/N]")
plt.plot(x_intersect, y_intersect, 'ro')
plt.show()
#P/W as function of WS
W_S_v=np.arange(1000,7000,100)
q=0.5*rho_OEI*v**2
```

```
AR=8              #Aspect Ratio
e0=0.7            #Oswald factor
C_L=(mTO_bas*g)/(q*S)
k=1/(pi*AR*e0)
CD0=(Epsilon*C_L)-1/(pi*AR*e0)*C_L**2
Epsilon_2OEI=((q_OEI*CD0)/W_S_v)+(k*W_S_v)/q_OEI
Epsilon_WS=((q*CD0)/W_S_v)+(k*W_S_v)/q
v_SR_WS=np.sqrt((2*1)/(cL_maxTO*rho_0)*W_S_v)
v2_min_WS=1.13*v_SR_WS
P0_WTO_WS=((q*CD0/W_S_v)+k*W_S_v/q)*kc*v/(eta_Prop*eta_Trans*TR_THR)
P0_WTO_OEI_WS=((q*CD0/W_S_v)+k*W_S_v/q)*kc*v_OEI/(eta_Prop*eta_Trans
    *TR_THR_OEI)*n_E/(n_E-1)
P0_WTO_2OEI_WS=(dh_ds+Epsilon_2OEI)*(v2_min_WS*n_E)/((1*0.8*
    eta_Trans)*(n_E-1))
P0_WTO_Ceiling_WS=((300/(v*196.85))+Epsilon_WS)*v/(0.9*0.99*0.9)
plt.ylim(0,40)
plt.xlabel("Wing Loading [N/m^2]")
plt.ylabel("Power to Weight Ratio [W/N]")
plt.grid(True)
plt.plot(W_S_v,P0_WTO_WS,label="Cruise",linewidth=2)
plt.plot(W_S_v,P0_WTO_OEI_WS,'--',label="Cruise OEI",linewidth=2)
plt.plot(W_S_v,P0_WTO_2OEI_WS,label="Climb OEI",linewidth=2)
plt.plot(W_S_v,P0_WTO_Ceiling_WS,label="Ceiling",linewidth=2)
plt.legend()
plt.show()

# ---FIND THE OPTIMAL DESIGN POINT (OP)---

max_P0_WTO = np.maximum.reduce([P0_WTO_WS, P0_WTO_OEI_WS,
    P0_WTO_2OEI_WS, P0_WTO_Ceiling_WS])
min_idx = np.argmin(max_P0_WTO)
W_S_opt = W_S_v[min_idx]
P0_WTO_min = max_P0_WTO[min_idx]

print(f"Optimal Wing Loading (W/S) : {W_S_opt} [N/m^2]")
print(f"Minimal Power to Weight ratio(P0/W_TO) : {P0_WTO_min:.4f} [W
    /N]")
```

18.4 Initial propulsion system sizing - Wing Loading - Stretch Variant

```
# Stretched Variant

#Calculations
W_S_Str=mTO_str*g/S                                        #Wing Loading streched variant [N/m ]
P0_WTO=(kc*Epsilon*v)/(eta_Prop*eta_Trans*TR_THR)          #Power to Weight Ratio [W/N]
P0_str=P0_WTO*(mOE_str*g)
#One engine inoperative
P0_OEI=(n_E-1)/n_E*P0_str
P0_WTO_OEI_str=(Epsilon_OEI*kc*v_OEI)/(TR_THR_OEI*eta_Trans*eta_Prop)*n_E/(n_E-1)

#Speeds
v_SR_str=sqrt((2*1)/(cL_maxTO*rho_0)*(mOE_str*g)/S)
v_LOF_str=1.08*v_SR_str
v2_min_str=1.13*v_SR_str
T_WTO_s1_str=1.15/(g*rho_0*cL_maxTO*s1*(1-1/(2*n_E)))*(mOE_str*g)/(S*1-0.03-0.13)
P0_WTO_S1_str=(T_WTO_s1_str*0.71*v_LOF)/(1*0.99*0.7)
T_WTO_s3_str = (np.sin(np.arctan(hscr_over_s3)) + E_ToOEI) / (1 - 1/n_E)
P0_WTO_S3_str=(T_WTO_s3_str*1/2*(v_LOF+v2_min_str))/(1*eta_Prop*eta_Prop)
P0_WTO_2OEI_str=(dh_ds+E_2OEI)*v2_min_str/(1*0.8*eta_Trans)*n_E/(n_E-1)
P0_WTO_Ceiling_str=(300/(v*196.85)+Epsilon)*v/(0.9*0.99*0.9)                    #300ft/min climb speed und horizontal speed Ma=0.6 but also in feet/min

#Result
print("Propulsion system sizing for strechted variante")
print("Power to weight Ratio =", round(float(P0_WTO),2), "[W/N]")
print("Power to weight Ratio OEI=", round(float(P0_WTO_OEI_str),2), "[W/N]")
print("Power to weight Ratio 2,OEI at climb=", round(float(P0_WTO_2OEI_str),2), "[W/N]")
print("Power to weight Ratio at ceiling =", round(float(P0_WTO_Ceiling_str),2), "[W/N]")
plt.xlabel("Meter")
plt.ylabel("Power to Weight Ratio")
plt.grid(True)

```

```
plt.plot(s1, P0_WTO_S1_str, label="P0_WTO_S1 Rolling distance for stretched variant", linewidth=2)
plt.plot(s1, P0_WTO_S3_str, '--', label="P0_WTO_S3 climbing distance for stretched variant", linewidth=2)
idx = np.argmin(np.abs(P0_WTO_S1_str - P0_WTO_S3_str))   # Find the crossing point of the two curves
plt.axvline(1300,color='black',label='Runway lenght')
plt.legend()
plt.ylim(0, 100)
x_intersect = s1[idx]
y_intersect = P0_WTO_S1_str[idx]
print("Required Power to Weight Ratio  for stretched variant to take of in given Runway lenght:", round(y_intersect,2),"[W/N]")
plt.plot(x_intersect, y_intersect, 'ro')
plt.show()

#P/W stretch Variant as function of WS
C_L_str=(mTO_str*g)/(q*S)
CD0_str=(Epsilon*C_L_str)-1/(pi*AR*e0)*C_L_str**2
Epsilon_2OEI_str=((q_OEI*CD0_str)/W_S_v)+(k*W_S_v)/q_OEI
Epsilon_WS_str=((q*CD0_str)/W_S_v)+(k*W_S_v)/q
P0_WTO_WS_str=((q*CD0_str/W_S_v)+k*W_S_v/q)*kc*v/(eta_Prop*eta_Trans*TR_THR)
P0_WTO_OEI_WS_str=((q*CD0_str/W_S_v)+k*W_S_v/q)*kc*v_OEI/(eta_Prop*eta_Trans*TR_THR_OEI)*n_E/(n_E-1)
P0_WTO_2OEI_WS_str=(dh_ds+Epsilon_2OEI_str)*(v2_min_WS*n_E)/((1*0.8*eta_Trans)*(n_E-1))
P0_WTO_Ceiling_WS_str=((300/(v*196.85))+Epsilon_WS_str)*v/(0.9*0.99*0.9)
plt.figure()
plt.ylim(0,40)
plt.xlabel("Wing Loading [N/m^2]")
plt.ylabel("Power to Weight Ratio [W/N]")
plt.grid(True)
plt.plot(W_S_v,P0_WTO_WS_str,label="Cruise",linewidth=2)
plt.plot(W_S_v,P0_WTO_OEI_WS_str,'--',label="Cruise OEI",linewidth=2)
plt.plot(W_S_v,P0_WTO_2OEI_WS_str,label="Climb OEI",linewidth=2)
plt.plot(W_S_v,P0_WTO_Ceiling_WS_str,label="Ceiling",linewidth=2)
plt.legend()
plt.show()

# ---FIND THE OPTIMAL DESIGN POINT (OP)---
max_P0_WTO_str = np.maximum.reduce([P0_WTO_WS_str,P0_WTO_OEI_WS_str,P0_WTO_2OEI_WS_str,P0_WTO_Ceiling_WS_str])
min_idx_str = np.argmin(max_P0_WTO_str)
W_S_opt_str = W_S_v[min_idx_str]
```

```python
P0_WTO_min_str = max_P0_WTO_str[min_idx_str]

print(f"Optimal wing loading (W/S) : {W_S_opt_str} [N/m^2]")
print(f"Minimal Power-to-weight ratio (P0/W_TO) : {P0_WTO_min_str:.4
    f} [W/N]")
```

18.5 Landing Requirements and Propeller Dimensioning

```
#Landing Requirements and Propeller Dimensioning Baseline Variante
cL_maxL=2.8                                          #Based on slide 57
WL_S=W_S*(1-k)                                       #Wing loading at landing
vS=np.sqrt((2*WL_S)/(rho_0*cL_maxL))
vL=1.13*vS                                           #Landing Speed
qL=0.5*rho_0*vL**2
Epsilon_L=(qL*CD0)/W_S+(k*W_S)/qL
h50=50
sL=(WL_S/(rho_0*cL_maxL))*((1.23**2-1.13**2)/(g*Epsilon_L)-1.13**2/-3)+h50/Epsilon_L #Landing distance
v50=1.23*vS*1.944                                    #Approch speed in knots

print("Landing Requirements Baseline Variante")
print("Total Landing distance= ",round(sL,2),"m")
print("Approch speed= ",round(v50,2),"knots")
K_Prop=0.49                                          #Based on slide 62
Pbr_E_base=eta_Trans*23*(mTO_bas*g)/(n_E*1000)       #Power per engine baseline variante in kW with an PO_WTO of 23W/N
d_prop_base=K_Prop*Pbr_E_base**(1/4)                 #Propeller Diameter Baseline Variante)
print("Propeller Diameter Baseline Variante= ",round(d_prop_base,2),"m")


#Landing Requirements and Propeller Dimensioning stretch Variante

WL_S_str=W_S_Str*(1-k)                               #Wing loading at landing
vS_str=np.sqrt((2*WL_S_str)/(rho_0*cL_maxL))
vL_str=1.13*vS_str                                   #Landing Speed
Epsilon_L_str=(qL*CD0)/W_S_Str+(k*W_S)/qL
sL_str=(WL_S_str/(rho_0*cL_maxL))*((1.23**2-1.13**2)/(g*Epsilon_L_str)-1.13**2/-3)+h50/Epsilon_L_str #Landing distance
v50_str=1.23*vS_str*1.944                            #Approch speed in knots

print("Landing Requirements stretch Variante")
print("Total Landing distance= ",round(sL_str,2),"m")
print("Approch speed= ",round(v50_str,2),"knots")

Pbr_E_stretch=eta_Trans*23*(mTO_str*g)/(n_E*1000)    #Power per engine stretch variante in kW with an PO_WTO of 23W/N
```

```
d_prop_stretch=K_Prop*Pbr_E_stretch**(1/4)            #Propeller Diameter stretch Variante)
print("Propeller Diameter stretch Variante= ",round(d_prop_stretch,2),"m")
```

18.6 Commonalities impacts

```
# Discussion about wings, engines and propellers commonality
##2026/01/10 - Selected values and commonalities impacts - Benoit Pereira

#Choices
W_S_str_com = 4500                                  #Chosen wing loading for stretched variant (N/m  )
P0_WTO_str_com = 23                                 #Chosen power to weight for stretched variant(W/N)



#Retroactive impacts on baseline parameters
W_S_bas_com = mTO_bas/mTO_str * W_S_str_com         #Effective wingloading  for baseline with common wings (N/m  )
P0_WTO_bas_com = mTO_str/mTO_bas * P0_WTO_str_com   #Effective power-to-weight ratio for baseline with common engines (W/N)

#Other commonalities impacts
D_prop = d_prop_stretch                             #Propeller commonality impact on propeller diameter (m)                 -> to meet stretched variant requirements
Pmec_Mot = P0_WTO_str_com*eta_Trans*mTO_str*g/1000  #Engine commonality impact on combined power output of the motors (kW) -> to meet stretched variant requirements



#Results
print("This gives the baseline an effective wingloading of", int(W_S_bas_com), "N/m  ")
print("This gives the baseline an effective power-to-weight ratio of ", int(P0_WTO_bas_com), "W/N")
print()
print("The common propeller diameter is", round(D_prop, 2), "m")
print("The common engines have a combined power output of", int(Pmec_Mot), "kW")
```

18.7 Hydrogen Systems - Fuel cells load optimizer

```
# Hydrogen systems
##2026/01/16 - Fuel Cells load optimizer - Benoit Pereira

from math import *
import matplotlib.pyplot as plt
import numpy as np

#Constants
etaInv = 0.99                   #Efficiency of the inverters
etaMot = 0.99                   #Efficiency of the electric motors
etaTrans = 0.98                 #Efficiency of the transmission
etaFC_anc = 0.93                #Efficiency of the fuel cells
    considering the ancillaries

etaFC_cool = 0.8                #Fraction of the heat produced by the
    fuel cells that must be dissipated by the cooling system

Pmr_FC_stack_100 = 9            #power-mass ratio for fuel cells stack
     at 100% load (kW/kg)
Pmr_FC_stack_33 = 3             #power-mass ratio for fuel cells stack
     at 33% load (kW/kg)
PVr_FC_stack_100 = 6            #power-volume ratio for fuel cells
    stack at 100% load (kW/L)
PVr_FC_stack_33 = 2             #power-volume ratio for fuel cells
    stack at 33% load (kW/L)

mPr_FC_anc = 0.17               #mass-power ratio for fuel cells
    ancillaries (at any load) (kg/kW)
VPr_FC_anc = 0.25               #volume-power ratio for fuel cells
    ancillaries (at any load) (L/kW)

mPr_FC_cool = 0.33              #mass-power ratio for fuel cells
    cooling (at any load) (kg/kW)
VPr_FC_cool = 0.50              #volume-power ratio for fuel cells
    cooling (at any load) (L/kW)



#Assumptions
TR_crs = 0.8                    #throttle ratio during cruise
TR_clmb = 0.9                   #throttle ratio  during climb
TR_to = 1                       #throttle ratio at take-off



#Calculations
```

```
##Powers
Pel_Mot = Pmec_Mot / etaMot
Pel_Inv = Pel_Mot / etaInv

Pel_nonProp = 0.0625 * Pel_Inv
Pel_FC_syst_crs = Pel_Inv * (TR_crs + 0.0625)
Pel_FC_syst_clmb = Pel_Inv * (TR_clmb + 0.0625)

##Fuel cells masses and volume vs Fuel Cells load
FCload_vec = np.linspace(0.1, 1.0, 100)

mFC_tot_vec = []
VFC_tot_vec = []
etaFC_stack_vec = []

for FCload in FCload_vec:

    ###Stack efficiency
    etaFC_stack = -0.213 * FCload * TR_to / TR_crs + 0.67
    etaFC_stack_vec.append(etaFC_stack)

    ###Cooling heat
    Qcool = etaFC_cool * Pel_Inv * (TR_to + 0.0625) * (1 /
       etaFC_stack - 1)

    ###Interpolated stack ratios
    mPr_FC_stack_load = 1 / (((Pmr_FC_stack_100 - Pmr_FC_stack_33) /
        (1 - 0.33)) * (FCload - 0.33) + Pmr_FC_stack_33)
    VPr_FC_stack_load = 1 / (((PVr_FC_stack_100 - PVr_FC_stack_33) /
        (1 - 0.33)) * (FCload - 0.33) + PVr_FC_stack_33)

    ###Volumes
    VFC_stack = Pel_FC_syst_crs * VPr_FC_stack_load
    VFC_anc = Pel_FC_syst_clmb * VPr_FC_anc
    VFC_cool = Qcool * VPr_FC_cool

    ###Masses
    mFC_stack = Pel_FC_syst_crs * mPr_FC_stack_load
    mFC_anc = Pel_FC_syst_clmb * mPr_FC_anc
    mFC_cool = Qcool * mPr_FC_cool

    ###Totals
    VFC_tot = VFC_stack + VFC_anc + VFC_cool
    mFC_tot = mFC_stack + mFC_anc + mFC_cool

    VFC_tot_vec.append(VFC_tot)
    mFC_tot_vec.append(mFC_tot)
```

```python

##Minima
###Convert to numpy arrays
mFC_tot_vec = np.array(mFC_tot_vec)
VFC_tot_vec = np.array(VFC_tot_vec)
etaFC_stack_vec = np.array(etaFC_stack_vec)

###Indices of minima
idx_m_min = np.argmin(mFC_tot_vec)
idx_V_min = np.argmin(VFC_tot_vec)

###Optimal FCload values
FCload_m_min = FCload_vec[idx_m_min]
FCload_V_min = FCload_vec[idx_V_min]

###Minimum values
mFC_min = mFC_tot_vec[idx_m_min]
VFC_min = VFC_tot_vec[idx_V_min]

###Corresponding efficiencies
eta_m_min = etaFC_stack_vec[idx_m_min]
eta_V_min = etaFC_stack_vec[idx_V_min]



#Results
print(f"Minimum FC system mass :")
print(f"  FCload = {FCload_m_min:.3f}")
print(f"  mFC_tot = {mFC_min:.2f} kg")
print(f"  etaFC_stack = {eta_m_min:.3f}\n")

print(f"Minimum FC system volume :")
print(f"  FCload = {FCload_V_min:.3f}")
print(f"  VFC_tot = {VFC_min:.2f} L")
print(f"  etaFC_stack = {eta_V_min:.3f}")



#Plot
fig, ax1 = plt.subplots(figsize=(8, 5))

##First axis for total FC mass and volume
ax1.plot(FCload_vec, VFC_tot_vec, label='Total FC volume (L)', linewidth=2)
ax1.plot(FCload_vec, mFC_tot_vec, label='Total FC mass (kg)', linewidth=2)
ax1.scatter(FCload_m_min, mFC_min, color='red', zorder=5)
```

```
ax1.scatter(FCload_V_min, VFC_min, color='blue', zorder=5)
ax1.set_xlabel('Fuel Cell Load')
ax1.set_xticks(np.arange(0.1, 1.01, 0.1))
ax1.set_ylabel('Mass / Volume')
ax1.set_ylim(0, 10000)
ax1.grid(True)

##Secondary axis for efficiency
ax2 = ax1.twinx()
ax2.plot(FCload_vec, etaFC_stack_vec, 'k--', label='FC stack
    efficiency')
ax2.set_ylabel('FC stack efficiency')
ax2.set_ylim(0, 1)

##Combined legend
lines1, labels1 = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax1.legend(lines1 + lines2, labels1 + labels2, loc='best')
plt.title('Fuel Cell system Mass, Volume and and stack efficiency vs
     FC Load')
plt.tight_layout()
plt.show()
```

18.8 Hydrogen Systems - Mass and volume estimation

```
# Hydrogen systems
##2025/12/12 - Hydrogen systems mass and volume estimation - Benoit Pereira

from math import *
import matplotlib.pyplot as plt
import numpy as np

#Constants and design specifications
d_LH2 = 0.0708                  #Density of LH2 (kg/L)
gravEff_LH2 = 0.5               #Proposed gravimetric efficiency of LH2 fuel tanks

mPr_Batt = 1/10                 #mass-power ratio for batteries (kg/kW)
VPr_Batt = 1/7.5                #volume-power ratio for batteries (L/kW)

mPr_Inv = 1/25                  #mass-power ratio for inverters (kg/kW)
VPr_Inv = 1/25                  #volume-power ratio for inverters (L/kW)

mPr_Mot = 1/12.5                #mass-power ratio for e-motors (kg/kW)
VPr_Mot = 1/25                  #volume-power ratio for e-motors (L/kW)



#Estimations/Assumptions/Design choices
BatPowerRatio = 0.1             #Batteries (booster pack) should provide an increase of power equal to 10% of the motors max power (for exceptional scenarios)
BatVolt = 3.7                   #Batteries (booster pack) voltage (V)
BatCrating = 40                 #Batteries (booster pack) C-rating (C)

Chosen_FCload = 0.45            #Fuel cells are oversized to be used at 45% charge during cruise



#Calculations
##LH2 tanks
VTanks = 1.072*mF_max/d_LH2                                         #tanks volume (1.072x larger than pure fuel volume) (L)
```

```
mTanks = mF_max*(1/gravEff_LH2-1)                                   #tanks mass (without fuel) (kg)
mTanks_Inst = mTanks*0.05                                           #Tanks installation systems mass (kg)
mFuelDist = mTanks*0.15                                             #Fuel distribution systems mass (kg)

##Powers
Pel_Mot = Pmec_Mot/etaMot                                           #Power input in the motors (= output of the inverters) (kW)
Pel_Inv = Pel_Mot/etaInv                                            #Power input in the inverters (= output of the fuel cells for propulsive purposes) (kW)

PelBat = BatPowerRatio*Pel_Mot                                      #Power output of the batteries (kW)

Pel_nonProp = 0.0625*Pel_Inv                                        #Power for non propulsive purposes (kW)
Pel_FC_syst_crs = Pel_Inv*(TR_crs+0.0625)                           #Power output from the fuel cells entire system during cruise (kW)
Pel_FC_syst_clmb = Pel_Inv*(TR_clmb+0.0625)                         #Power output from the fuel cells entire system during climb (kW)
Pel_stack = Pel_FC_syst_crs/etaFC_anc                               #Power output of the fuel cells stack (during cruise) (kW)

etaFC_stack = -0.213*Chosen_FCload* TR_to/TR_crs +0.67              #Stack efficiency (depends of the load)
Qcool = etaFC_cool*Pel_Inv*(TR_to+0.0625)*(1/etaFC_stack-1)         #Heat flow handled by fuel cells cooling

##Fuels cells
mPr_FC_stack_load = 1 / (((Pmr_FC_stack_100 - Pmr_FC_stack_33)/(1-0.33)) * (Chosen_FCload-0.33) + Pmr_FC_stack_33)          #linearly interpolated mass-power ratio for (oversized) fuel cells stack (kg/kW)
VPr_FC_stack_load = 1 / (((PVr_FC_stack_100 - PVr_FC_stack_33)/(1-0.33)) * (Chosen_FCload-0.33) + PVr_FC_stack_33)          #linearly interpolated volume-power ratio for (oversized) fuel cells stack (L/kW)

VFC_stack = Pel_FC_syst_crs*VPr_FC_stack_load                       #Fuel cells stack volume (L)
mFC_stack = Pel_FC_syst_crs*mPr_FC_stack_load                       #Fuel cells stack mass (kg)
VFC_anc = Pel_FC_syst_clmb*VPr_FC_anc                               #Fuel cells ancillaries volume (L)
```

```
mFC_anc = Pel_FC_syst_clmb*mPr_FC_anc                                #Fuel cells ancillaries mass (kg)
VFC_cool = Qcool*VPr_FC_cool                                         #Fuel cells cooling volume (L)
mFC_cool = Qcool*mPr_FC_cool                                         #Fuel cells cooling mass (kg)

VFC_tot = VFC_stack + VFC_anc + VFC_cool                             #Total fuel cells system volume (L)
mFC_tot = mFC_stack + mFC_anc + mFC_cool                             #Total fuel cells system mass (kg)

##Batteries
VBatt = PelBat*mPr_Batt                                              # Batteries volume (L)
mBatt = PelBat*mPr_Batt                                              # Batteries mass (kg)
BatCap = PelBat/BatVolt*BatCrating                                   # Batteries capacity (kAh)

##Inverters
VInv = Pel_Inv*VPr_Inv                                               # Inverters volume (L)
mInv = Pel_Inv*mPr_Inv                                               # Inverters mass (kg)

##E_motors
VMot = Pel_Mot*VPr_Mot                                               #total motors volume (include gearbox) (L)
mMot = Pel_Mot*mPr_Mot                                               #total motors mass (include gearbox) (kg)
mMot_cool = 0.2*(mInv+mMot)                                          # Cooling systems mass (for motors and inverters) (kg)

##Total
mHEF = mTanks+mTanks_Inst+mFuelDist+mFC_tot+mBatt+mInv+mMot+mMot_cool            #Total mass of the hydrogen systems + motors (everything computed in this part)



#Results
print(" === LH2 tanks === ")
print("LH2 tanks volume =", int(VTanks), "L")
print("LH2 tanks mass =", int(mTanks), "kg")
print("LH2 tanks installation systems mass =", int(mTanks_Inst), "kg")
```

```
print("LH2 distribution systems mass =", int(mFuelDist), "kg")
print()
print(" === Fuel cells === ")
print("Fuel cells stack volume =", int(VFC_stack), "L")
print("Fuel cells stack mass =", int(mFC_stack), "kg")
print("Fuel cells ancillaries volume =", int(VFC_anc), "L")
print("Fuel cells ancillaries mass =", int(mFC_anc), "kg")
print("Fuel cells cooling volume =", int(VFC_cool), "L")
print("Fuel cells cooling mass =", int(mFC_cool), "kg")
print()
print("Total fuel cells system volume =", int(VFC_tot), "L")
print("Total fuel cells system mass =", int(mFC_tot), "kg")
print()
print("Fuel cells stack efficiency =", round(float(etaFC_stack), 2))
print()
print(" === Batteries === ")
print("Batteries volume =", int(VBatt), "L")
print("Batteries mass =", int(mBatt), "kg")
print("Batteries capacity =", int(BatCap), "kAh")
print()
print(" === Inverters === ")
print("Inverters volume =", int(VInv), "L")
print("Inverters mass =", int(mInv), "kg")
print()
print(" === E-Motors (total) === ")
print("E-motors volume =", int(VMot), "L")
print("E-motors mass =", int(mMot), "kg")
print("E-motors cooling systems mass =", int(mMot_cool), "kg")
print()
print(" === Total === ")
print("Total HEF systems + motors mass =", int(mHEF), "kg")
```

18.9 Hydrogen Systems - CAD help

```
# Support for HEF systems CAD (gives dimensions for specified lengths/diameters)
##2026/01/06 - HEF systems CAD help - Benoit Pereira

from math import *

###E-motors and inverters are represented by cylinders
###LH2 tanks are represented by cylinders with spherical ends
###Fuel cells and batteries are represented by cuboids

#Variable parameters
Dtanks = 1.15                           #Diameter of the tanks (m)                          (value to modify in order to test different configurations)

Dmot = 0.5                              #Diameter of the engines (m)                          (value to modify in order to test different configurations)

Dinv = 0.75                             #Diameter of the inverters (m)                          (value to modify in order to test different configurations)

H_FC = 0.40                             #Height of the fuel cells (m)                          (value to modify in order to test different configurations)
W_FC = 1.35                             #Width of the fuel cells (m)                          (value to modify in order to test different configurations)

H_batt = 0.40                           #Height of the batteries (m)                          (value to modify in order to test different configurations)
W_batt = 1.35                           #Width of the batteries (m)                          (value to modify in order to test different configurations)



#Calculations
##Tanks
Stanks=pi*(Dtanks/2)**2                 #Cross-section area (m  )
V1tank = VTanks/1000/2                  #Volume of a single tank (there are 2 in total) (m  )
V_sph_tank = 4/3*pi*(Dtanks/2)**3       #Volume of the spherical parts (ends of the tank) (m  )
```

```
V_cyl_tank = V1tank-V_sph_tank        #Volume of the cylindrical part (m  )
L_cyl_tank = V_cyl_tank/Stanks        #Length of the cylindrical part (m)
L_tank = L_cyl_tank + Dtanks          #Total length of the tanks (m)

##Fuel cells
L_FC = (VFC_tot/1000)/(H_FC*W_FC)     #Length of the fuel cells (m)

##Batteries
L_batt = (VBatt/1000)/(H_batt*W_batt) #Length of the batteries (m)

##Inverters
Sinv=pi*(Dinv/2)**2                   #Cross-section area (m  )
V1Inv = VInv/1000/2                   #Volume of a single inverter (there are 2 in total) (m  )
L_1Inv = V1Inv/Sinv                   #Length of the cylinder (m)

##E-motors
Smot=pi*(Dmot/2)**2                   #Cross-section area (m  )
V1Mot = VMot/1000/2                   #Volume of a single E_motor (there are 2 in total) (m  )
L_1Mot = V1Mot/Smot                   #Length of the cylinder (m)



#Results
print(" === E_motors === ")
print("Chosen diameter is ", Dmot, "m")
print("Length of 1 engine is ", round(L_1Mot, 2), "m")
print()
print(" === Inverters === ")
print("Chosen diameter is ", Dinv, "m")
print("Length of 1 inverter is ", round(L_1Inv, 2), "m")
print()
print(" === Batteries === ")
print("Length of the batteries cuboid is", round(L_batt, 2), "m")
print()
print(" === Fuel cells === ")
print("Length of the fuel cells cuboid is", round(L_FC, 2), "m")
print()

if L_cyl_tank >= 0:
  print(" === Tanks === ")
  print("Chosen diameter is ", Dtanks, "m")
  print("Lenth of 1 spherical end is", round(Dtanks/2, 3), "m")
```

```
  print("Lenth of the cylindrical part is", round(L_cyl_tank, 2), "m
      ")
  print("Total length of 1 tank is", round(L_tank, 2), "m")
  print()
  print("Dont forget we need 2 of these pods !")

else:
  print("Diameter is too big even with just a spherical tank")
```

18.10 Configuration - Wing Dimensioning

```
# Wing dimensionning
##2025/12/13 - Wing dimensionning - L na Aveline

#Constants and design specifications
gamma = 1.4                                                    #Ratio of specific heat for air
P_290 = 30700                                                  #Pressure at flight level 290
M_MO = 0.57                                                    #Maximum operation Mach number
k_n = 1.3                                                      #Factor for maneuvers and gust
k_av = 1.1                                                     #Factor for increase of the local lift coefficient
k_trim = 1.1                                                   #Factor for trimming



#Assumptions/Design choices
AR=8                                                           #Aspect ratio (assumption made for initial propulsion sizing)
TR=0.45                                                        #Taper ratio [0.3;0.5]
phi25 = 0.0                                                    #No sweep at 25% chord



#Calculations
##Wings Geometry
S_wing = (mTO_str*g)/W_S_str_com                               #Wing area [m  ]
b_wing = sqrt(AR*S_wing)                                       #Span [m]
c_root = (2*S_wing)/(b_wing*(1+TR))                            #Root chord [m]
c_tip = TR*c_root                                              #Tip chord [m]
c_mid = S_wing/b_wing                                          #Mid chord [m]

##Wing geometry for double trapezoidal
###Point and chord at kink point (Assumption: kink point at the engine level)
y_kink = 0.4*(b_wing/2)
```

```
c_kink = c_root - ((c_root - c_tip)/(b_wing/2)*y_kink)
###Taper Ratios
TR_in = c_kink/c_root
TR_out = c_tip/c_kink
###Mean Aerodynamic Chord
c_mac_in = (2/3)*c_root*((1+TR_in+TR_in**2)/(1+TR_in))
c_mac_out = (2/3)*c_kink*((1+TR_out+TR_out**2)/(1+TR_out))
###MAC positions
y_mac_in = (y_kink/6)*((1+2*TR_in)/(1+TR_in))
y_mac_out = ((((b_wing/2)-y_kink)/6)*((1+2*TR_out)/(1+TR_out)))+
    y_kink
###Surfaces
S_in = 2*y_kink*((c_root+c_kink)/2)
S_out = 2*((b_wing/2)-y_kink)*((c_kink+c_tip)/2)
###global MAC
c_mac = (S_in*c_mac_in+S_out*c_mac_out)/S_wing
y_mac = (S_in*y_mac_in+S_out*y_mac_out)/S_wing

##Lift coefficients
cl_bas= (W_S_bas_com)*(2/(gamma*P_290*M_MO**2))                #Lift
    coefficient for baseline
cl_BO_bas = k_n*k_av*k_trim*cl_bas                             #
    Buffet Onset Lift Coefficient for baseline
cl_DD_bas = k_av*k_trim*cl_bas                                 #Drag
    Divergence Lift coefficient for baseline

cl_str= (W_S_str_com)*(2/(gamma*P_290*M_MO**2))                #Lift
    coefficient for stretched variant
cl_BO_str = k_n*k_av*k_trim*cl_str                             #
    Buffet Onset Lift Coefficient for stretched variant
cl_DD_str = k_av*k_trim*cl_str                                 #Drag
    Divergence Lift coefficient for stretched variant




#Results
print(" === Wing geometry === ")
print("Wing surface =", round(float(S_wing), 2), " m  ")
print("Wing span =", round(float(b_wing),2), "m")
print("Mid chord lenght =", round(float(c_mid), 2), "m")
print("Root chord lenght =", round(float(c_root), 2), "m")
print("Tip chord lenght =", round(float(c_tip), 2), "m")
print()
print(" === Geometry for double trapezoidal === ")
print("kink point is in y =", round(float(y_kink), 2))
print("Chord at kink point =", round(float(c_kink), 2), "m")
```

```
print()
print("Mean aerodynamic chord for the inner panel is in y =", round(
    float(y_mac_in), 2),)
print("Mean aerodynamic chord for the inner panel =",round(float(
    c_mac_in), 2), "m")
print()
print("Mean aerodynamic chord for the outer panel is in y =", round(
    float(y_mac_out), 2),)
print("Mean aerodynamic chord for the outer panel =",round(float(
    c_mac_out), 2), "m")
print()
print("Total surface of inner panels =", round(float(S_in), 2), "m
    ")
print("Total surface of outer panels =", round(float(S_out), 2), "
    m  ")
print()
print("Mean aerodynamic chord is in y =", round(float(y_mac), 2))
print("Mean aerodynamic chord =",round(float(c_mac), 2), "m")
print()
print(" === Lift coefficients === ")
print("Lift coefficient for baseline = ", round(float(cl_bas), 2))
print("Buffet Onset lift coefficient for baseline =", round(float(
    cl_BO_bas), 3))
print("Drag Divergence lift coefficient for baseline =", round(float
    (cl_DD_bas), 3))
print()
print("Lift coefficient for stretched variant = ", round(float(
    cl_str), 2))
print("Buffet Onset lift coefficient for stretched variant =", round
    (float(cl_BO_str), 3))
print("Drag Divergence lift coefficient for stretched variant =",
    round(float(cl_DD_str), 3))
print()



#Sketch of double trapezoid wing
##Sweep at 25% chord
phi25_rad = np.deg2rad(phi25)

def section_coords(y, c):
    x_25 = y * np.tan(phi25_rad)
    x_le = x_25 - 0.25 * c
    x_te = x_25 + 0.75 * c
    return x_le, x_te, x_25

##Sections
```

```
###Root
y_r = 0.0
x_le_r, x_te_r, x_25_r = section_coords(y_r, c_root)
###Kink
x_le_k, x_te_k, x_25_k = section_coords(y_kink, c_kink)
###Tip
y_t = b_wing / 2
x_le_t, x_te_t, x_25_t = section_coords(y_t, c_tip)

##Plotting
plt.figure(figsize=(10, 4))
###Leading edge
plt.plot([y_r, y_kink], [x_le_r, x_le_k], 'k')
plt.plot([y_kink, y_t], [x_le_k, x_le_t], 'k')
###Trailing edge
plt.plot([y_r, y_kink], [x_te_r, x_te_k], 'k')
plt.plot([y_kink, y_t], [x_te_k, x_te_t], 'k')
###Other half
plt.plot([-y_r, -y_kink], [x_le_r, x_le_k], 'k')
plt.plot([-y_kink, -y_t], [x_le_k, x_le_t], 'k')
plt.plot([-y_r, -y_kink], [x_te_r, x_te_k], 'k')
plt.plot([-y_kink, -y_t], [x_te_k, x_te_t], 'k')
###Sweep line
plt.plot([0, y_t], [x_25_r, x_25_t], 'b--', label='25% chord line')
plt.plot([0, -y_t], [x_25_r, x_25_t], 'b--')
###Root chord
plt.plot([y_r, y_r], [x_le_r, x_te_r], 'k--')
###Kink chord
plt.plot([y_kink, y_kink], [x_le_k, x_te_k], 'k--')
plt.plot([-y_kink, -y_kink], [x_le_k, x_te_k], 'k--')
###Tip chord
plt.plot([y_t, y_t], [x_le_t, x_te_t], 'k--')
plt.plot([-y_t, -y_t], [x_le_t, x_te_t], 'k--')

##Mean aerodynamic chord
x_25_mac = y_mac * np.tan(phi25_rad)
x_le_mac = x_25_mac - 0.25 * c_mac
x_te_mac = x_25_mac + 0.75 * c_mac
plt.plot([y_mac, y_mac], [x_le_mac, x_te_mac], 'r--', linewidth=2,
    label='MAC')
plt.plot([-y_mac, -y_mac], [x_le_mac, x_te_mac], 'r--', linewidth=2)

##Aerodynamic center
x_ac = x_25_r
y_ac = 0.0
plt.plot(y_ac, x_ac, 'ro', label='Aerodynamic center')

```

```python

plt.xlabel("Spanwise coordinate y [m]")
plt.ylabel("Chordwise coordinate x [m]")
plt.title("Double Trapezoidal Wing Planform")
plt.axis("equal")
plt.grid(True)
plt.legend()
plt.gca().invert_xaxis()
plt.gca().invert_yaxis()
plt.show()
```

18.11 Configuration - Empennage Dimensioning

```
# Empennage dimensionning
##16/12/2025 - Empennage dimensioning -

from math import *
import matplotlib.pyplot as plt
import numpy as np

#Coefficients according to slide 18 raymer
c_h= 0.8                                    #Horizontal tailplane volume coefficient
c_v=0.07                                    #Vertical tailplane volume coefficient
lf= 20                                      #fuselage lenght in m. Just an assumption now
lh=0.5*lf                                   #distance between aerodynamic center of wing and tailplane in m
lv=lh



#Calculations
Sh=(c_h*S_wing*c_mid)/lh                    #Horizontal tailplane area in m
Sv=(c_v*S_wing*b_wing)/lv                   #Vertical tailplane area in m



#Results
print("Horizontal tailplane area =", round(float(Sh),2), "m  ")
print("Vertical tailplane area =", round(float(Sv),2), "m  ")
```

18.12 Configuration - Fuselage Dimensioning

```
from math import *
import matplotlib.pyplot as plt
import numpy as np

#Constants/given values

SW=20                                       #Seat width including armrests (in)
WA=20                                       #Width of the passenger aisle (in)

roh_cargo=160                               #Average density of cargo (kg/m  )
roh_Bag=165                                 #Average density of Luggage (kg/m  )

GSV=0.028                                   #Galley storage volume needed per passenger (m  )



#Design choices/estimations/assumptions

SPR=4                                       #Seats per Row
SP=36                                       #Seat pitch (in)

lc_bas=14                                   #length of the cabin for the base variante according to slide 12 (m)
lc_str=22                                   #length of the cabin for the stretch variante according to slide 12 (m)



#Calculations
##Cross-section
d=(SPR*SW+WA)*2.54/100                      #inner diameter of the fuselage (m)
tw=0.02*d+0.0254                            #wall thickness of the fuselage (m)
tf=1.5*tw                                   #floor thickness (m)
D=d+2*tw                                    #outer diameter of the fuselage (m)

#X-axis lengths
l_cockpit=1.5*D                             #Length of the cockpit 1.3-1.7 (m)
```

```
l_tail=3.0*D                                #Length of the tail 2.7-3.2 (m)
l_ges_bas=l_tail+l_cockpit+lc_bas           #total length of the base aircraft (m)
l_ges_str=l_tail+l_cockpit+lc_str           #total length of the stretched aircraft (m)

##cargo
V_Cargo=mFrght/roh_cargo                    #Volume of the cargo (m  )
V_Bag_bas=(mLugg*NbPass_bas)/roh_Bag        #Volume of the Luggage for the base variante (m  )
V_Bag_str=(mLugg*NbPass_str)/roh_Bag        #Volume of the Luggage for the stretched variante (m  )

##Galleys
V_Gal_bas=GSV*NbPass_bas                    #Total galleys storage volume for the base variante (m  )
V_Gal_str=GSV*NbPass_str                    #Total galleys storage volume for the stretched variante (m  )



#Results
print(" === Cross-section === ")
print("Inner diameter =", round(float(d),2), "m")
print("Outer diameter =", round(float(D),2), "m")
print("Wall thickness =", round(float(tw),2), "m")
print("Floor thickness =", round(float(tf),2), "m")
print()
print(" === X-axis lengths === ")
print("Cockpit length", round(float(l_cockpit),2), "m")
print("Base variant cabin length =", round(float(lc_bas),2), "m")
print("Stretched variant cabin length =", round(float(lc_str),2), "m")
print("Tail length =", round(float(l_tail),2), "m")
print()
print("Base variant total length =", round(float(l_ges_bas),2), "m")
print("Stretched variant total length =", round(float(l_ges_str),2), "m")
print()
print(" === Cargo === ")
print("Cargo volume", round(float(V_Cargo),3), "m  ")
print("Base variant lugage volume =", round(float(V_Bag_bas),3), "m  ")
print("Stretched variant lugage volume =", round(float(V_Bag_str),3), "m  ")
```

```
print()
print(" === Galleys === ")
print("Base variant galleys storage volume =", round(float(V_Gal_bas
    ),3), " m  ")
print("Stretched variant galleys storage volume =", round(float(
    V_Gal_str),3), " m  ")
```